TEMPLATE

KEY PROJECT INFORMATION & PROJECT DESIGN DOCUMENT (PDD)

PUBLICATION DATE **14.10.2020**

VERSION **v. 1.2**

RELATED SUPPORT

- **TEMPLATE GUIDE Key Project Information & Project Design Document v.1.2**

This document contains the following Sections

Key Project Information

0 – Description of project

0 - Application of approved Gold Standard Methodology (ies) and/or demonstration of SDG Contributions

0 – Duration and crediting period

0 – Summary of Safeguarding Principles and Gender Sensitive Assessment

0 – Outcome of Stakeholder Consultations

Appendix 1 – Safeguarding Principles Assessment (mandatory)

0 - Contact information of Project participants (mandatory)

0 - LUF Additional Information (project specific)

0 - Summary of Approved Design Changes (project specific)

KEY PROJECT INFORMATION

GS ID of Project	GS-11214
Title of Project	Pedro Corto Solar
Time of First Submission Date	18/07/2021
Date of Design Certification	04/07/2023
Version number of the PDD	V 6
Completion date of version	27/11/2023
Project Developer	Soventix Caribbean SRL
Project Representative	Alfonso Rodríguez
Project Participants and any communities involved	Soventix Caribbean SRL Irradiasol Dominicana SRL
Host Country (ies)	Dominican Republic
Activity Requirements applied	☐ Community Services Activities ☒ Renewable Energy Activities ☐ Land Use and Forestry Activities/Risks & Capacities ☐ N/A
Scale of the project activity	☐ Micro scale ☐ Small Scale X Large Scale
Other Requirements applied	NA
Methodology (ies) applied and version number	ACM0002 Grid-connected electricity generation from renewable sources Version 20.0
Product Requirements applied	☒ GHG Emissions Reduction & Sequestration ☐ Renewable Energy Label N/A
Project Cycle:	X Regular ☐ Retroactive

Table 1 – Estimated Sustainable Development Contributions

Sustainable Development Goals Targeted	SDG Impact (defined in B.6.)	Estimated Annual Average	Units or Products
SDG 7 Affordable and Clean Energy	Quantity of net electricity generation supplied by the project plant to the grid in year y	151,262 MWh	
SDG 9 Industry, Innovation and Infrastructure	Total number of renewable energy plants in the Dominican energy mix.	3	
SDG13 Climate Action	ER expected in the crediting period	94,024GS VERs/year	
SDG 8 Decent Work and Economic Growth	Total number of jobs created through the project	Approx. 500 jobs (during construction phase) and 100 jobs (during operation and maintenance phase)	

SECTION A. DESCRIPTION OF PROJECT

A.1 Purpose and general description of project

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Pedro Corto Solar consists of an 82.5 MW photovoltaic solar installation located in Pedro Corto, province San Juan, Dominican Republic. The project is located near the border to Haiti, one of the least developed areas of the country. The proposed project activity will supply an average of 82.5 MW to the SENI (Sistema Eléctrico Nacional Interconectado), National Interconnected Electric System (the grid) avoiding part of the electricity generated by the grid-connected plants. This is the current scenario (scenario existing prior to the implementation of the project) and it is also the baseline scenario (scenario that would occur without the implementation of the project). This means that the baseline scenario would be that all the electricity delivered to the grid by the project would have been generated by the grid-connected power plants and by the addition of new generation resources.

A.1.1. Eligibility of the project under Gold Standard

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Pedro Corto Solar project meets the eligibility criteria meets the eligibility criteria as per section 3.1.1 of the GS4GG Principles & Requirements document version 1.2, including paragraphs 2 and 3 of the Renewable Energy Activity Requirement document version 1.4, and GHG Emissions Reductions & Sequestration Product Requirements as shown below:

As per section 3.1.1 of the GS4GG Principles & Requirements document version 1.2:

a) Types of Project: Eligible projects shall include physical action/implementation on the ground. Pre-identified eligible project types are identified in the Eligibility Principles and Requirements section. Pedro Corto Solar shall generate and deliver renewable energy (solar photovoltaic) to the national grid of the Dominican Republic. The project will use the CDM approved methodology ACM0002, Version 21.0, an eligible methodology under Gold Standard, hence the condition is satisfied.

b) Location of Project: Projects may be located in any part of the world. The project is located in Pedro Corto, San Juan Province, Dominican Republic, 18◦51’40.74’’N

71◦23’08.32’’W, hence the condition is satisfied.

c) Project Area, Project Boundary and Scale: The Project Area and Project Boundary shall be defined. Projects may be developed at any scale although certain rules, requirements and limitations may apply under specific Activity Requirements, Impact Quantification Methodologies and Products Requirements. In order to avoid double counting the Project shall not be included in any other voluntary or compliance standards program unless approved by Gold Standard (for example through dual certification). Also, if the Project Area overlaps with that of another Gold Standard or other voluntary or compliance standard program of a similar nature, the project shall demonstrate that there is no double counting of impacts at design and performance certification (for example use of similar technology or practices through which the potential arises for double counting or misestimation of impacts amongst projects). According to the methodology ACM0002, the spatial extent of the project boundary includes the project power plant/unit and all power plants/units connected physically to the electricity system that the proposed project power plant is connected to.

   Additionally, according to the “Tool to calculate the emission factor for an electricity system” (Tool 7 Version 7.0), a grid/project electricity system is defined by the spatial extent of the power plants that are physically connected through transmission and distribution lines to the project activity and that can be dispatched without significant transmission constraints, hence the condition is satisfied.

   In this case, the project activity is connected to the National Interconnected Electrical System (SENI). Thus, Pedro Corto Solar site is connected to the SENI and included in the project boundary, hence the condition is satisfied.

   The project has not been registered and is not seeking registration under any other GHG programs. The project has no intentions to generate any other form of GHG related environmental credit or for GHG reduction of emissions or removal claimed under the Gold Standard Certification, hence the condition is satisfied.

d) Host Country Requirements: There are two main laws which define the regulatory framework of the electricity generation sector in Dominican Republic in general, and to promote renewable energy in particular. These are briefly described below. The General Electricity Act (created by the law 125-01 of 2001) and its modification with the law 186-07 of August 2007. The main issues defined by the law are the following:

- Assurance that at least 20 % of all electricity trading is done on the spot market.
- Limiting distribution company ownership of generating plants to not more than 15% of peak load in the interconnected system (renewable energy sources are exempt from this rule).
- Regulation of electricity tariffs for public-grid customers with maximum connected loads of 2 MW as long as the customers do not enter into direct contracts with the suppliers.
- Electricity companies which intend to operate an electricity generation business above 2 MW must request a concession for the operation of electricity works.
- Regulation of transit tariffs for the use of transmission and distribution of facilities.
- Provision of preferential treatment to companies that generate electricity from renewable energy sources with regard to sales and load distribution. - if prices and conditions are otherwise identical.
- Exemption of companies generating electricity using renewable energy sources from national and local taxes for five years.

The Law on Incentives for the Development of Renewable Energy Sources and Special Regimes, No. 57-07 and its Implementation Regulation. The law looks to promote the energy diversity of the country in terms of self-sufficiency of strategic inputs, meaning fuel and non-conventional energy, reducing its dependency on imported fossil fuels, stimulating private investment to develop projects from renewable energy sources, and mitigate the negative environmental impacts of fossil fuel energy operations.

This law promotes wind farms with a capacity of up to 50 MW, mini hydropower plants of up to 5 MW, PV installations of all sizes, concentrating solar thermal power stations of up to 120 MW, biomass power stations, and ocean power plants. The law also promotes technologies for solar heat generation and refrigeration. The incentives include the following:

- 100% exemption over import duties for equipment, machinery and accessories required for renewable energy production.
- 100% exemption over sales tax, for all above mentioned equipment.
- Reduction to a fixed 5% on the tax over foreign financed interest

payments, modifying Art. 306 of the Dominican Tax Code for the beneficiaries of this new law.

- Owners or renters of family homes and commercial or industrial establishments who shift to renewable energy systems for their private consumption are given a tax credit equal to 40% of the capital cost of the equipment purchased. Consumers will be able to discount the tax credit from their income tax for the next three years at a proportion of 33.33% per year.
- This law also calls for the creation of a CO_2 emissions bond market under the platform of the Kyoto Protocol, regulated by the Ministry of Natural Resource's Mechanism of Clean Development.

Since the project activity is a photovoltaic solar installation, there is no legal requirement given by Law No. 57-07 to incorporate any other non-conventional renewable energy (NCRE). The proposed project activity is fully in compliance with all mandatory laws and regulations; hence the condition is satisfied.

e) Contact Details: As part of the Project Documentation the Project Developer shall provide (i) name and (ii) contact details of 7 all Project Participants; AND in case of an organization (iii) the legal registration details and (iv) documentation by the governing jurisdiction that proves that the entity is in good standing (defined as being a legal or other appropriate entity registered in or allowed to operate within the required jurisdiction and with no evidence of insolvency or legal/criminal notices placed against it or any of its Directors). Gold Standard retains the right (at its own discretion) to refuse use of the Standard where reputational concerns are highlighted.

   As shown in Appendix 2, the contact information of Project participants Irradiasol Dominicana SRL and Soventix Caribbean SRL, hence the condition is satisfied.

f) Legal Ownership: Full and uncontested legal ownership of any Products that are generated under Gold Standard Certification, (for example carbon credits) shall be demonstrated. Where such ownership is transferred from project beneficiaries this must be demonstrated transparently and with full, prior and informed consent (FPIC). Note that for certain Project types there is a requirement for full and uncontested legal land title/tenure to be demonstrated. These are contained within specific Activity or Product Requirements. All projects shall immediately report to Gold Standard any land title/tenure disputes arising.

Legal Ownership: The owner of the project is the company Irradiasol Dominicana, S.R.L., which is an organized commercial society existing under the laws of the Dominican Republic, attached you can find the Certificate of Commercial Register Limited Liability Company Commercial Register No.156708SD, hence the condition is satisfied.

g) Other Rights: As well as legal title and ownership, the Project Developer shall also demonstrate where required uncontested legal rights and/or permissions concerning changes in use of other resources required to service the Project (for example, access rights, water rights etc.). Any known disputes or contested rights must be declared immediately to Gold Standard by the Project Developer and resolved prior to further project implementation in affected areas.

   The proposed project activity is fully in compliance with all mandatory laws and regulations; hence the condition is satisfied.

h) Official Development Assistance (ODA) Declaration: All Project Developers applying for project activities located in a country named by the OECD Development Assistance Committee's ODA recipient list and seeking Gold Standard Certification for carbon credits shall declare the Official Development Assistance (ODA) support. The Project Developer shall follow the GHG Emissions Reduction & Sequestration Product Requirements and submit the declaration at the time of Design Certification.

   The Dominican Republic is an Upper Middle-Income Country according to the 2021 DAC List of ODA recipients. Irradiasol Dominicana SRL declares that it does not receive or benefit from ODA, and therefore, attached you will find the Official Development Assistance Declaration completed and signed, hence the condition is satisfied.

As per paragraph 2 of the Renewable Energy Activity Requirement document version 1.4:

a) Projects shall generate and deliver energy services (e.g., mechanical work/electricity/heat) from non-fossil and renewable energy sources. Pedro Corto Solar shall generate and deliver renewable energy (solar) to the national grid of the Dominican Republic; hence the condition is satisfied.
b) Projects shall comprise of renewable energy generation units, such as photovoltaic, tidal/wave, wind, hydro, geothermal, waste to energy and renewable biomass, that are supplying energy to a national or a regional grid; or supplying energy to an identified consumer facility via national/regional grid through a contractual agreement such as wheeling. Pedro Corto Solar shall generate and deliver renewable energy to the national grid of the Dominican Republic; hence the condition is satisfied.
c) Any Project supplying electricity to a minigrid shall refer to Community Services

Activity Requirements. The condition is not applicable as the project activity is not supplying to a minigrid.

d) Projects generating on-site energy for captive consumption at an industrial facility shall refer to the requirements in this document. The condition is not applicable as the project does not generate on-site energy for captive consumption.

e) New Gold Standard Verified Emission Reductions (GS VER) or Gold Standard labels for Certified Emission Reductions (GS CER), Renewable Energy projects connected to national, or a regional electricity grid must be located in a;

   a. Least Developed Country (LDC), Small Island Developing State (SIDS)or a Land Locked Developing Country (LLDC) or
   b. Low Income and Low Middle-income country where the penetration level of the proposed Renewable Energy Technology type is less than5% of the total grid installed capacity, at the time of the firstsubmission to preliminary review.

   Renewable Energy projects connected to national, or a regional electricity gridare ineligible for GS VERs/CERs, if located in:

   - An Upper Middle-Income Country or High-Income Country or
   - SIDS and LLDC, defined as a High- Income Country

   As this project is connected to the national grid in the Dominican Republic and the Dominican Republic is an Upper-Middle Income country and not a High-Income Country (https://data.worldbank.org/country/dominican-republic) located in a SIDS (https://www.un.org/ohrlls/content/list-sids), according to the latest classification by income from the World Bank, the project is deemed eligible, hence the condition is satisfied.

f) Grid Connected off-shore wind projects and waste to energy projects that involve utilization of landfill gas/biogas to electricity generation with or without thermal energy production are exempted from eligibility requirement outlined in paragraph above. The condition is not applicable as the project is a solar energy project.

As per paragraph 3 of the Renewable Energy Activity Requirement document version 1.4:

a) Project Type: As mentioned above, the project belongs to a pre-identified eligible project type. The project is located in the Dominican Republic, a SIDS and is classified as an Upper-Middle Income Country by World Bank according to their latest classification of countries as per income level. Therefore, the project is deemed eligible, hence the condition is satisfied.

b) Location of Project: Projects may be located in any part of the world. The project is located in Pedro Corto, San Juan Province, Dominican Republic, 18◦51’40.74’’N 71◦23’08.32’’W

c) Project area, boundary and scale: Pedro Corto Solar is classified as a large scale and defined as so, as per the Renewable Energy Activity Requirement document (Version 1.4) that deems large scale projects as projects with a maximum output capacity above 15 MW, hence the condition is satisfied.

d) Suppressed demand: Pedro Corto Solar does not have a suppressed demand as it is not a micro-scaleor small-scale project, hence the condition is satisfied.

e) Stacking: Pedro Corto Solar will not be stacking as it is only pursuing the Gold Standard certification for GS VERs. Perdro Corto Solar is not pursuing any other certification to obtain carbon credits including Verra, CDM, etc., as it is only applying to Gold Standard to register the project, hence the condition is satisfied.

As per the GHG Emissions Reductions & Sequestration Product Requirements, Pedro Corto Solar is located in the Dominican Republic which is a country that does not have a mandatory operational scheme to reduce GHG emissions in any form and does not have the possibility to trade emissions that include the scope of the proposed project.

In the case of eligible greenhouse gases, CO_2 is the GHG gas that will be displaced due to the project activity. Pedro Corto Solar is a solar plant connected to the National grid that will supply renewable energy making it eligible for GSVERs. As the Dominican Republic is named by the OECD Development Assistance Committee´s ODA Recipient, Irradiasol Dominicana SRL has signed and submitted the ODA Declaration.

Note: The Dominican Republic is a recipient of the funds given by the OECD, butthis project will not receive any funds and therefore Irradiasol Dominicana SRL signed adeclaration stating that it will not receive any of these funds for the project. The declaration is signed and uploaded.

A.1.2. Legal ownership of products generated by the project and legal rights to alter use of resources required to service the project

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The owner of the project is the company Irradiasol Dominicana, SRL, which is an organized commercial society and existing under the laws of the Dominican Republic, with its registered office located in Calle Marginal de la Av. Núñez de Cáceres No. 366, Edificio Corporativo NC Piso 13, Santo Domingo de Guzmán, Distrito Nacional, Dominican Republic, registered under the Commercial Register with No. 156708SD and under the National Taxpayer´s Registry with No. 1-31-9069-4, whose social object is the generation of energy produced with non-conventional and/or renewable sources of energy.

A.2 Location of project

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Dominican Republic, Pedro Corto, San Juan province.

The following maps show the location of the project:

Figure 1. Location of Pedro Corto Solar in the San Juan province.

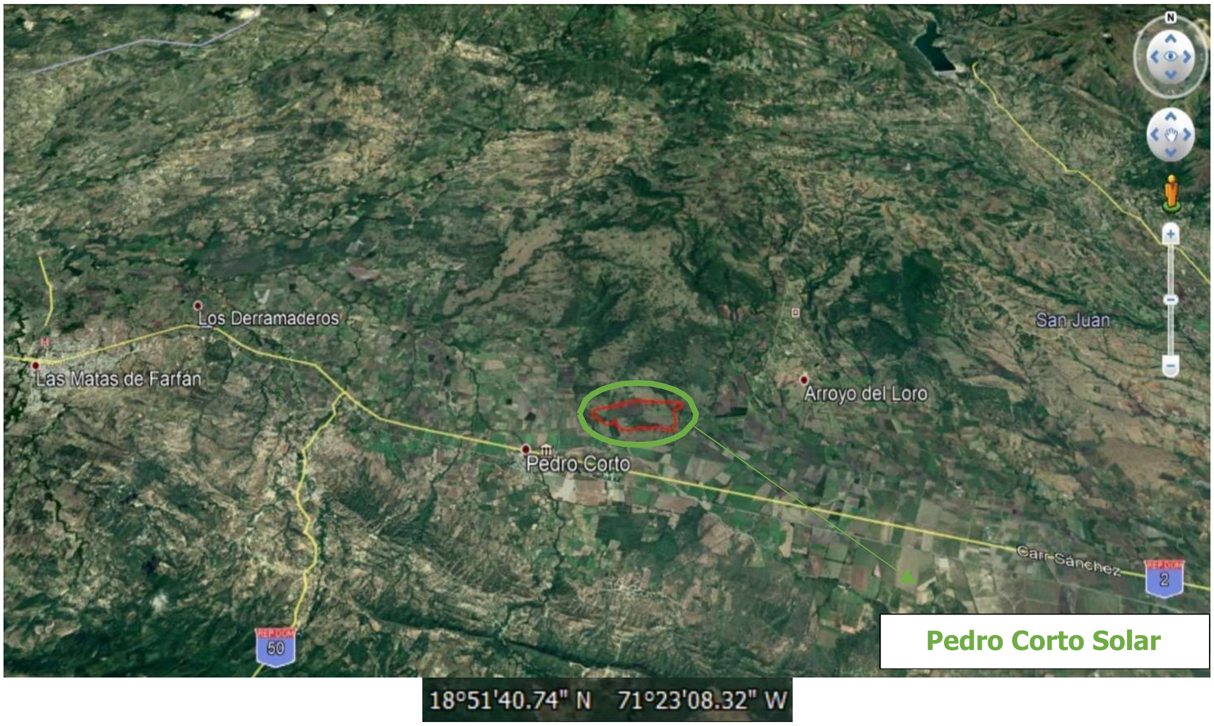

Figure 2. Pedro Corto Solar satellite view.

A.) Technologies and/or measures

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The technologies and measures to be implemented by the project are subject to change during course of the project since photovoltaic technology is changing all the time. The technology inside the solar panels is becoming more efficient and therefore the description below of equipment, systems and facilities may be subject to change.

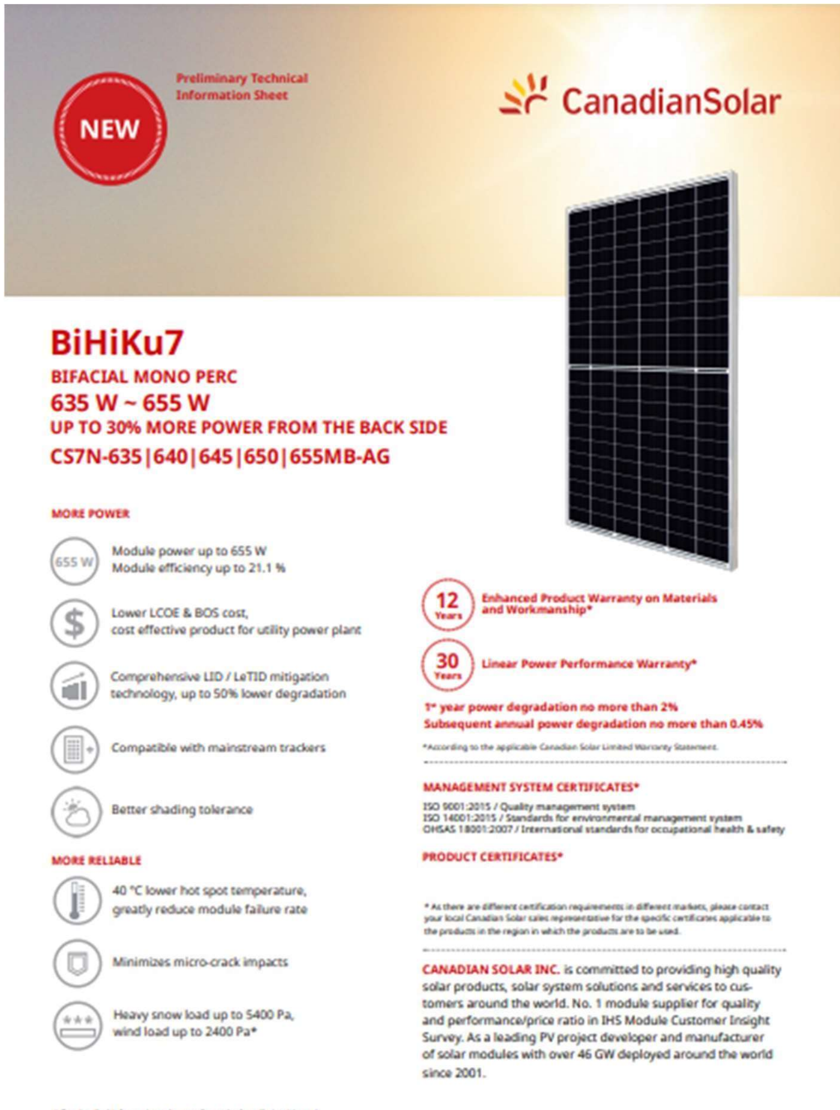
Note: The plausible modifications will not imply any significant financial variation or ER estimate. While the efficiency of the photovoltaic equipment has reached a tipping point, the availability of the proposed equipment at the time of construction may vary due to market availability restrictions.

The 125,952 photovoltaic modules will be distributed in two types of tables with fixed tilt structure. The distribution of the tables is the following:

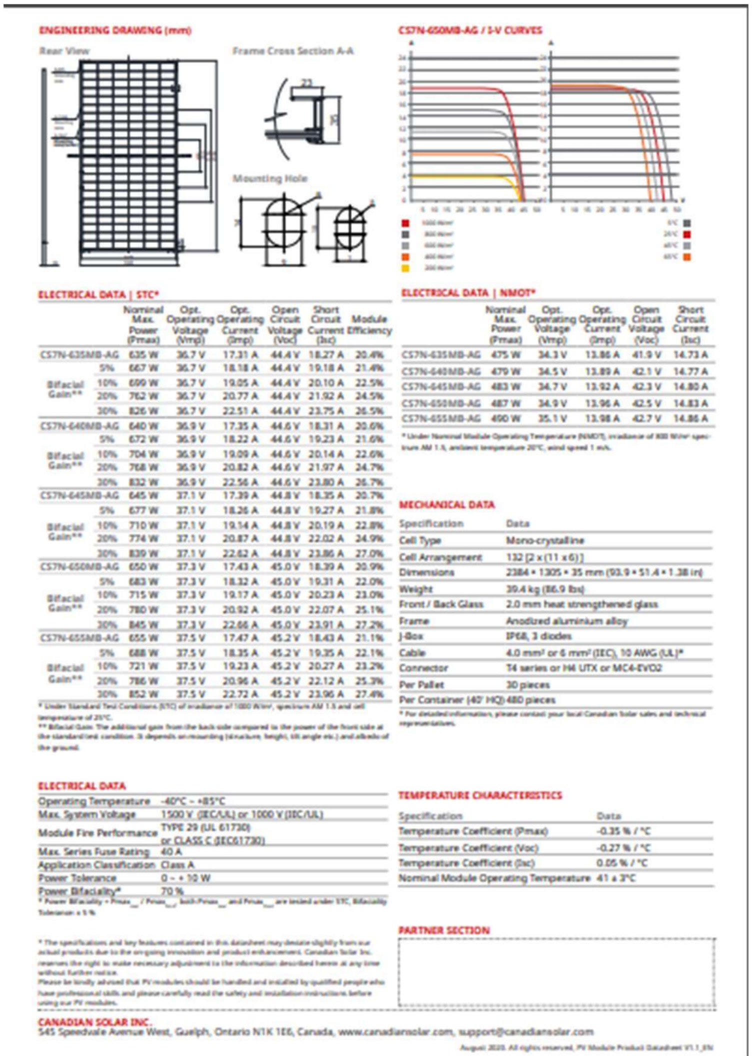
a) Approximately 88% of the tables (866 tables approximately) will have 128 panels per table, 4 strings with 32 panels per string. These tables are located in the inner part of the project.
b) Approximately 12% (118 tables approximately) of the tables will have 64 panels per table, 2 strings with 32 panels each per table-. This tables are located in the outer part of the project, suffering more wind loads.

The modules include connection boxes for outdoor, which include by-pass to avoid a possible interruption of the electrical circuit inside the module, as a consequence of partial shading of any cell (in these situations reverse current is produced that may break the diode product of over intensity).

The following is a list of the main characteristics of the modules that will be used in the project:



NEW
Preliminary Technical Information Sheet
CanadianSolar
BiHiKu7
BIFACIAL MONO PERC
635 W ~ 655 W
UP TO 30% MORE POWER FROM THE BACK SIDE
CS7N-635|640|645|650|655MB-AG
MORE POWER
655 W
Module power up to 655 W
Module efficiency up to 21.1 %
Lower LCOE & BOS cost,
cost effective product for utility power plant
Comprehensive LID / LeTID mitigation technology, up to 50% lower degradation
Compatible with mainstream trackers
Better shading tolerance
MORE RELIABLE
40 °C lower hot spot temperature, greatly reduce module failure rate
Minimizes micro-crack impacts
Heavy snow load up to 5400 Pa, wind load up to 2400 Pa*
* For detailed information, please refer to the Installation Manual.
12 Years
Enhanced Product Warranty on Materials and Workmanship*
30 Years
Linear Power Performance Warranty*
1st year power degradation no more than 2%
Subsequent annual power degradation no more than 0.45%
*According to the applicable Canadian Solar Limited Warranty Statement.
MANAGEMENT SYSTEM CERTIFICATES*
ISO 9001:2015 / Quality management system
ISO 14001:2015 / Standards for environmental management system
OHSAS 18001:2007 / International standards for occupational health & safety
PRODUCT CERTIFICATES*
* As there are different certification requirements in different markets, please contact your local Canadian Solar sales representative for the specific certificates applicable to the products in the region in which the products are to be used.
CANADIAN SOLAR INC. is committed to providing high quality solar products, solar system solutions and services to customers around the world. No. 1 module supplier for quality and performance/price ratio in IHS Module Customer Insight Survey. As a leading PV project developer and manufacturer of solar modules with over 46 GW deployed around the world since 2001.



ENGINEERING DRAWING (mm)

Rear View — Frame Cross Section A-A — Mounting Hole

CS7N-650MB-AG / I-V CURVES

ELECTRICAL DATA | STC*

		Nominal Max. Power (Pmax)	Opt. Operating Voltage (Vmp)	Opt. Operating Current (Imp)	Open Circuit Voltage (Voc)	Short Circuit Current (Isc)	Module Efficiency
CS7N-635MB-AG		635 W	36.7 V	17.31 A	44.4 V	18.27 A	20.4%
Bifacial Gain**	5%	667 W	36.7 V	18.18 A	44.4 V	19.18 A	21.4%
	10%	699 W	36.7 V	19.05 A	44.4 V	20.10 A	22.5%
	20%	762 W	36.7 V	20.77 A	44.4 V	21.92 A	24.5%
	30%	826 W	36.7 V	22.51 A	44.4 V	23.75 A	26.5%
CS7N-640MB-AG		640 W	36.9 V	17.35 A	44.6 V	18.31 A	20.6%
Bifacial Gain**	5%	672 W	36.9 V	18.22 A	44.6 V	19.23 A	21.6%
	10%	704 W	36.9 V	19.09 A	44.6 V	20.14 A	22.6%
	20%	768 W	36.9 V	20.82 A	44.6 V	21.97 A	24.7%
	30%	832 W	36.9 V	22.56 A	44.6 V	23.80 A	26.7%
CS7N-645MB-AG		645 W	37.1 V	17.39 A	44.8 V	18.35 A	20.7%
Bifacial Gain**	5%	677 W	37.1 V	18.26 A	44.8 V	19.27 A	21.8%
	10%	710 W	37.1 V	19.14 A	44.8 V	20.19 A	22.8%
	20%	774 W	37.1 V	20.87 A	44.8 V	22.02 A	24.9%
	30%	839 W	37.1 V	22.62 A	44.8 V	23.86 A	27.0%
CS7N-650MB-AG		650 W	37.3 V	17.43 A	45.0 V	18.39 A	20.9%
Bifacial Gain**	5%	683 W	37.3 V	18.32 A	45.0 V	19.31 A	22.0%
	10%	715 W	37.3 V	19.17 A	45.0 V	20.23 A	23.0%
	20%	780 W	37.3 V	20.92 A	45.0 V	22.07 A	25.1%
	30%	845 W	37.3 V	22.66 A	45.0 V	23.91 A	27.2%
CS7N-655MB-AG		655 W	37.5 V	17.47 A	45.2 V	18.43 A	21.1%
Bifacial Gain**	5%	688 W	37.5 V	18.35 A	45.2 V	19.35 A	22.1%
	10%	721 W	37.5 V	19.23 A	45.2 V	20.27 A	23.2%
	20%	786 W	37.5 V	20.96 A	45.2 V	22.12 A	25.3%
	30%	852 W	37.5 V	22.72 A	45.2 V	23.96 A	27.4%

ELECTRICAL DATA | NMOT*

	Nominal Max. Power (Pmax)	Opt. Operating Voltage (Vmp)	Opt. Operating Current (Imp)	Open Circuit Voltage (Voc)	Short Circuit Current (Isc)
CS7N-635MB-AG	475 W	34.3 V	13.86 A	41.9 V	14.73 A
CS7N-640MB-AG	479 W	34.5 V	13.89 A	42.1 V	14.77 A
CS7N-645MB-AG	483 W	34.7 V	13.92 A	42.3 V	14.80 A
CS7N-650MB-AG	487 W	34.9 V	13.96 A	42.5 V	14.83 A
CS7N-655MB-AG	490 W	35.1 V	13.98 A	42.7 V	14.86 A

MECHANICAL DATA

Specification	Data
Cell Type	Mono-crystalline
Cell Arrangement	132 [2 x (11 x 6)]
Dimensions	2384 • 1305 • 35 mm (93.9 • 51.4 • 1.38 in)
Weight	39.4 kg (86.9 lbs)
Front / Back Glass	2.0 mm heat strengthened glass
Frame	Anodized aluminium alloy
J-Box	IP68, 3 diodes
Cable	4.0 mm² or 6 mm² (IEC), 10 AWG (UL)*
Connector	T4 series or H4 UTX or MC4-EVO2
Per Pallet	30 pieces
Per Container (40' HQ)	480 pieces

ELECTRICAL DATA

Operating Temperature	-40°C ~ +85°C
Max. System Voltage	1500 V (IEC/UL) or 1000 V (IEC/UL)
Module Fire Performance	TYPE 29 (UL 61730) or CLASS C (IEC61730)
Max. Series Fuse Rating	40 A
Application Classification	Class A
Power Tolerance	0 ~ + 10 W
Power Bifaciality*	70 %

TEMPERATURE CHARACTERISTICS

Specification	Data
Temperature Coefficient (Pmax)	-0.35 % / °C
Temperature Coefficient (Voc)	-0.27 % / °C
Temperature Coefficient (Isc)	0.05 % / °C
Nominal Module Operating Temperature	41 ± 3°C

PARTNER SECTION

CANADIAN SOLAR INC.
545 Speedvale Avenue West, Guelph, Ontario N1K 1E6, Canada, www.canadiansolar.com, support@canadiansolar.com

The following table shows the main technical characteristics of the PV module:

PV Module (Custom Parameters Definition)	
Manufacturer	Canadian Solar Inc.
Model	CS7N-655MB-AG 1500V
Unit Nom. Power	655 Wp
Number of PV modules	125,952 units
Nominal (STC)	82.50 MWp
Modules	3936 Strings x 32 In series

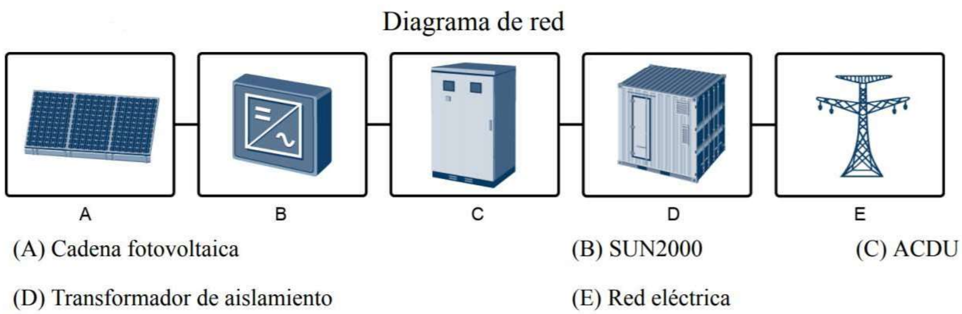
One of the most important elements in a photovoltaic installation is the metallic support structure, in which the solar modules are fixed, and the optimal inclination is given for a perfect utilization of the solar radiation, this way optimizing the efficiency of the photovoltaic plant.

For Pedro Corto Solar, the tables with metallic structures will be used from the manufacturer Schletter, PV Powerway or similar. These structures are made from prefabricated galvanized steel, designed for this specific end and with high resistance to wind and certain charges. Equally, all of the mechanic elements for fixation and screws are galvanized steel to prevent corrosion. The fixation of the structures and safeguards of the photovoltaic modules must resist winds up to 240 km/h, also, some overloads in accordance with the international applicable regulations, in which case is justified by the estimate of the charges and structural analysis. The metallic parts of the structure will be connected to the grounding point of the installation.

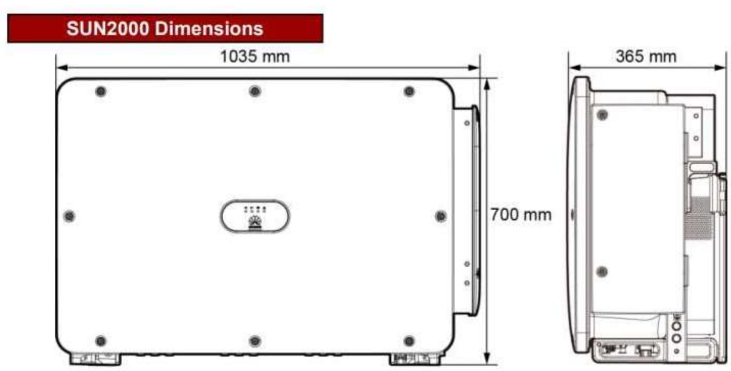
For the installation, 396 HUAWEI inverters will be used, model HUAWEI SUN2000-185kTL-H, which is a three-phase photovoltaic inverter connected to the electrical grid that converts the DC power generated by the photovoltaic chain to AC power that feeds the electrical grid with this power output.

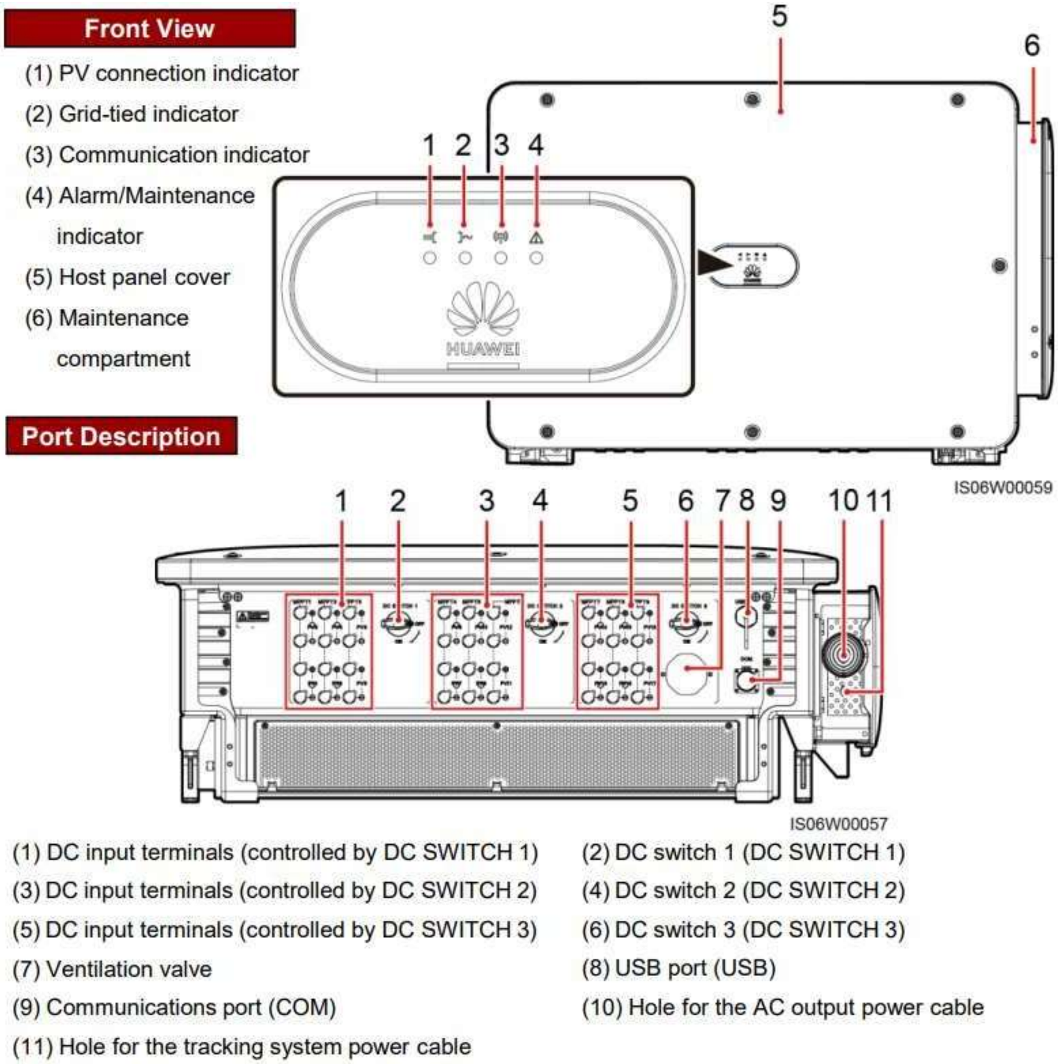
The inverters that are going to be used are designed to inject the power by the photovoltaic modules to the conventional electrical grid. Its main mission is to guarantee the quality of the energy released to the grid, also to centralize a series of protections for the maintenance workers and the owner of the installation. The inverter is tasked to convert the generated energy in the photovoltaic field in DC power to an 800 V AC power and synchronize the frequency with the grid.

The inverter meets all the established protections in the current legislation, especially with the electromagnetic compatibility standards and the technical requirements established by the legislation for electrical installations connected to the grid.



The characteristics and technical specifications of the inverter are as follows:




Front View
(1) PV connection indicator
(2) Grid-tied indicator
(3) Communication indicator
(4) Alarm/Maintenance indicator
(5) Host panel cover
(6) Maintenance compartment
HUAWEI
IS06W00059
Port Description
IS06W00057
(1) DC input terminals (controlled by DC SWITCH 1)
(2) DC switch 1 (DC SWITCH 1)
(3) DC input terminals (controlled by DC SWITCH 2)
(4) DC switch 2 (DC SWITCH 2)
(5) DC input terminals (controlled by DC SWITCH 3)
(6) DC switch 3 (DC SWITCH 3)
(7) Ventilation valve
(8) USB port (USB)
(9) Communications port (COM)
(10) Hole for the AC output power cable
(11) Hole for the tracking system power cable


SUN2000-185KTL-H1
Smart String Inverter
9
MPP Trackers
99.0%
Max. Efficiency
String-level
Management
Smart I-V Curve
Diagnosis Supported
MBUS
Supported
Fuse Free
Design
Surge Arresters for
DC & AC
IP66
Protection
Efficiency Curve
Circuit Diagram

SUN2000-185KTL-H1

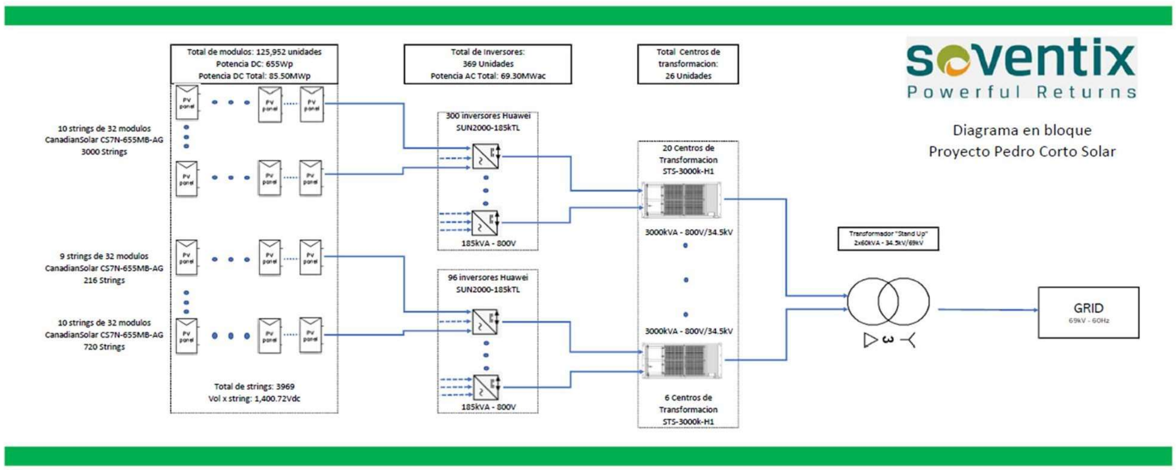
Technical Specifications

Efficiency	
Max. Efficiency	99.03%
European Efficiency	98.69%
Input	
Max. Input Voltage	1,500 V
Max. Current per MPPT	26 A
Max. Short Circuit Current per MPPT	40 A
Start Voltage	550 V
MPPT Operating Voltage Range	500 V ~ 1,500 V
Nominal Input Voltage	1,080 V
Number of Inputs	18
Number of MPP Trackers	9
Output	
Nominal AC Active Power	175,000 W @40°C
Max. AC Apparent Power	185,000 VA
Nominal Output Voltage	800 V, 3W + PE
Rated AC Grid Frequency	50 Hz / 60 Hz
Nominal Output Current	126.3 A @40°C
Max. Output Current	134.9 A
Adjustable Power Factor Range	0.8 LG ... 0.8 LD
Max. Total Harmonic Distortion	< 3%
Protection	
Input-side Disconnection Device	Yes
Anti-islanding Protection	Yes
AC Overcurrent Protection	Yes
DC Reverse-polarity Protection	Yes
PV-array String Fault Monitoring	Yes
DC Surge Arrester	Type II
AC Surge Arrester	Type II
DC Insulation Resistance Detection	Yes
Residual Current Monitoring Unit	Yes
Communication	
Display	LED Indicators, Bluetooth/WLAN + APP
USB	Yes
MBUS	Yes
RS485	Yes
General	
Dimensions (W x H x D)	1,035 x 700 x 365 mm (40.7 x 27.6 x 14.4 inch)
Weight (with mounting plate)	84 kg (185.2 lb.)
Operating Temperature Range	-25°C ~ 60°C (-13°F ~ 140°F)
Cooling Method	Smart Air Cooling
Max. Operating Altitude without Derating	4,000 m (13,123 ft.)
Relative Humidity	0 ~ 100%
DC Connector	Staubli MC4 EVO2
AC Connector	Waterproof Connector + OT/DT Terminal
Protection Degree	IP66
Topology	Transformerless
Standard Compliance (more available upon request)	
Certificates	EN 62109-1/-2, IEC 62109-1/-2, EN 50530, IEC 62116, IEC 60068, IEC 61683, IEC 61727, IEC 62910, P.O. 12.3, RD 1699, RD 661, RD 413, RD 1565, RD 1663, ABNT NBR 16149, ABNT NBR 16150, ABNT NBR IEC 62116

The following table shows the main technical characteristics of the Inverter:

Inverter (Custom Parameters Definition)	
Manufacturer	Huawei Technologies
Model	SUN2000-185KTL-H1-40C-Preliminary-v0.1
Unit Nom. Power	175 kVA
Number of Inverters	396 Unit
Total Power	69,300 kVA
Operating Voltage	600-1500 V
Max. Power (=>30°C)	185 kVA
Pnom Ratio (DC:AC)	1.19

The single-line diagram of the project represents graphically the equipment and elements that comprise Pedro Corto Solar.



The technical document "Access Study to the High-Tension Electric Grid of the SENI of the Photovoltaic Solar Park Pedro Corto of Irradiasol Dominicana SRL", has the objective to analyze the insertion of a non-conventional renewable energy park in the Pedro Corto Municipality, province San Juan, Dominican Republic. In this study, the capacity of the grid to absorb the generated potency by this new project is explored.

Pedro Corto Solar will contribute to the sustainable development (SDG) of the Caribbean region and the Dominican Republic as follows:

1. The project will replace conventional fossil fuel-based power generation with clean renewable solar power generation and will decrease the emissions to the atmosphere of sulphur oxides (SO_x), nitrogen oxides (NO_x), carbon monoxide (CO), and carbon dioxide (CO2), particulate matter and other pollutants associated with the combustion of fossil fuels.
2. The project will reduce the use of fossil fuels in thermal power plants, and therefore will help decrease the dependency of the country on fossil fuels and other non-renewable resources, which are limited in supply.

3. The project activity will help the Dominican Republic achieve its emission reduction goals by 2030 and increase the renewable energy mix of the National Energy Matrix.
4. The project activity will generate employment opportunities for the local community during the construction phase and once the project is implemented for the operation and maintenance activities.
5. The project activity will contribute to the technology transfer and technological self-reliance through the capacity building of employees. In addition, the project will generate experience in the solar sector, strengthening the institutional capacities and increasing investor´s confidence in this technology.
6. The project will contribute to the local development and income level of the community by creating new direct and indirect income sources.

A.4 Scale of the project

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The project is a grid connected large scale renewable energy production on a Small Island Developing State. The project is considered a large-scale project as small-scale projects are defined in accordance with GS project standard for project activities as a renewable energy project with a maximum output capacity of 15 MW (e) or 45 MW (th) and all projects exceeding the small-scale thresholds are defined as large scale. Pedro Corto Solar is an 82.5 MW project.

A.5 Funding sources of project

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The project will be funded by the investors of Irradiasol Dominicana SRL through a proper debt/equity. While today the source of financing has not been defined, typical financing for these type of projects in the Dominican Republic is as follows:

- Debt/Equity ratio: 70/30.
- Tenor/crediting period: 14 years.
- All in interest rate: 6.0% Dollars (All-in interest rate of 6.0% refers to annual interest rate.)

SECTION B. APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES) AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

B.1. Reference of approved methodology (ies)

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The project activity is developed in accordance with the CDM approved methodology ACM0002 "Grid-connected electricity generation from renewable sources" (Version 21.0) from UNFCCC.

According to this methodology, the grid emission factor is calculated using the "Tool to calculate the emission factor for an electricity system" (Tool 07, Version 7.0) and the project additionality is demonstrated using "Tool for the demonstration and assessment of additionality" (Version 07.0.0, Tool 01).

B.2. Applicability of methodology (ies)

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Applicability condition 1 of ACM0002:
The methodology ACM0002 (Version 21.0) is applicable to grid-connected renewable energy powergeneration project activities that:

1. Install a Greenfield power plant;
2. Involve a capacity addition to (an) existing plant(s);
3. Involve a retrofit of (an) existing operating plants/units;
4. Involve a rehabilitation of (an) existing plant(s)/unit(s); or
5. Involve a replacement of (an) existing plant(s)/unit(s).

In this case, the proposed project activity of Pedro Corto Solar involves a grid-connected renewable energy power generation through the installation of a photovoltaic solar plant (a Greenfield power plant). Pedro Corto Solar is a greenfield power plant as per definition of the Methodology ACM0002, as it is a new renewable energy plant that is constructed and operated at a site where no renewable energy power plant was operated prior to the implementation of the project activity. Pedro Corto Solar is a new solar plant that is constructed and operated at a site where no plant of the same type was operated prior to the implementation of the project activity. Below pictures can demonstrate that the project is a Greenfield power plant and there was no renewable energy project prior to the implementation of Pedro Corto Solar. The pictures are part of the Environmental Impact Study.

Foto 1. *Vista en perspectiva hacia el Norte del área del proyecto (tomada 30 de octubre 2020).*

Figure 3. Perspective view of the north area of the project (Picture taken on October 30th, 2020)

Foto 2.3.3-3. *Suelos limo-arcillosos en el área del proyecto (tomada 30 de octubre 2020).*

Figure 4. Lime clayish soils in the area of the project (Picture taken on October 30th, 2020)

Foto 2.4.1.1-1. Vegetación tipo potrero con árboles dispersos (tomada 30 de octubre 2020).

Figure 5. Vegetation type meadow with scattered trees (Picture taken on October 30th, 2020)

The applicability conditions of the methodology ACM0002 are presented below together with an explanation on why each condition is met by the project activity.

Applicability Conditions	Justification by project developer
1. The project activity may include renewable energy power plant/unit of one of the following types: hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit;	The proposed project activity involves renewable energy power generation through the installation of a solar power plant. Thus, this condition is met.

2. In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project Activity	The proposed project activity involves the installation of a Greenfield power plant. Thus, this condition is not applicable.
3. In case of Greenfield project activities applicable under paragraph 5 (a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the renewable energy project activity (e.g. by referring to feasibility studies or investment decision documents); **4.** The BESS should be charged with electricity generated from the associated renewable energy power plant(s). Only during exigencies 2 may the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions following the requirements under section 5.4.4 below. The charging using the grid or using fossil fuel electricity generator should not amount to more than 2 per cent of the electricity generated by the project renewable energy plant during a monitoring period. During the time	The proposed project activity involves the installation of a Greenfield power plant that does not include BESS. Thus, this condition is not applicable. The condition is not applicable as the project is a solar power project and not an integrated BESS.

periods (e.g. week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period.	

5. In case of hydro power plants, one of the following conditions shall apply: (a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or (b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation (7), is greater than 4 W/m_2; or (c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m_2; or (d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m_2, all of the following conditions shall apply: (i) The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m_2; (ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; (iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m_2 shall be:	The proposed project activity involves the installation of a solar power plant. Thus, this condition is not applicable.

a. Lower than or equal to 15 MW; and b. Less than 10 per cent of the total installed capacity of integrated hydro power project.	
6. In the case of integrated hydro power projects, project proponent shall: (a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or (b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum of five years prior to the implementation of the CDM project activity.	The proposed project activity involves the installation of a solar power plant. Thus, this condition is not applicable.

7. The methodology is not applicable to: (a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; (b) Biomass fired power plants/units.	The proposed project activity does not involve switching from fossil fuels to renewable energy sources at the project site or the installation of biomass fired power plants. Thus, this condition is not applicable.
8. In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is "the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance".	The proposed project activity involves the installation of a Greenfield power plant. Thus, this condition is not applicable

<table>
<tr><th colspan="2">Source</th><th>GHGs</th><th>Included?</th><th>Justification/Explanation</th></tr>
<tr><td rowspan="4">Baseline scenario</td><td rowspan="4">CO_2 emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity.</td><td>CO_2</td><td>Yes</td><td>Main emission source</td></tr>
<tr><td>CH_4</td><td>No</td><td>Minor emission source</td></tr>
<tr><td>N_2O</td><td>No</td><td>Minor emission source</td></tr>
<tr><td>Other</td><td>No</td><td>NA</td></tr>
<tr><td rowspan="11">Project scenario</td><td rowspan="4">For dry or flash steam geothermal power plants, emissions of CH_4 and CO_2 from non-condensable gases contained in geothermal steam.</td><td>CO_2</td><td>No</td><td rowspan="4">The project activity involves the installation of a solar power plant.</td></tr>
<tr><td>CH_4</td><td>No</td></tr>
<tr><td>N_2O</td><td>No</td></tr>
<tr><td>Other</td><td>No</td></tr>
<tr><td rowspan="4">For binary geothermal power plants, fugitive emissions of CH_4 and CO_2 from non-condensable gases contained in geothermal steam.</td><td>CO_2</td><td>No</td><td rowspan="4">The project activity involves the installation of a solar power plant.</td></tr>
<tr><td>CH_4</td><td>No</td></tr>
<tr><td>N_2O</td><td>No</td></tr>
<tr><td>Other</td><td>No</td></tr>
<tr><td>For binary geothermal power plants, fugitive emissions of hydrocarbons such as n-butane and isopentane (working fluid) contained in the heat exchangers.</td><td>Low GWP hydrocarbon/ refrigerant.</td><td>No</td><td>The project activity involves the installation of a solar power plant.</td></tr>
<tr><td>For hydro power plants, emissions of CH_4 from the reservoir.</td><td>CO_2
CH_4
N_2O
Other</td><td>No
No
No
No</td><td>The project activity involves the installation of a solar power plant.</td></tr>
<tr><td>CO_2 emissions from combustion of fossil fuels for electricity generation in solar thermal power plants and geothermal power plants.</td><td>CO_2
CH_4
N_2O
Other</td><td>No
No
No
No</td><td>The project activity involves the installation of a solar power plant. As can be seen in Figures 3, 4 and 5 above there are no diesel power plants in the project boundary. Electricity generation for the project will be obtained from the solar power plant.</td></tr>
</table>

B.). Project boundary

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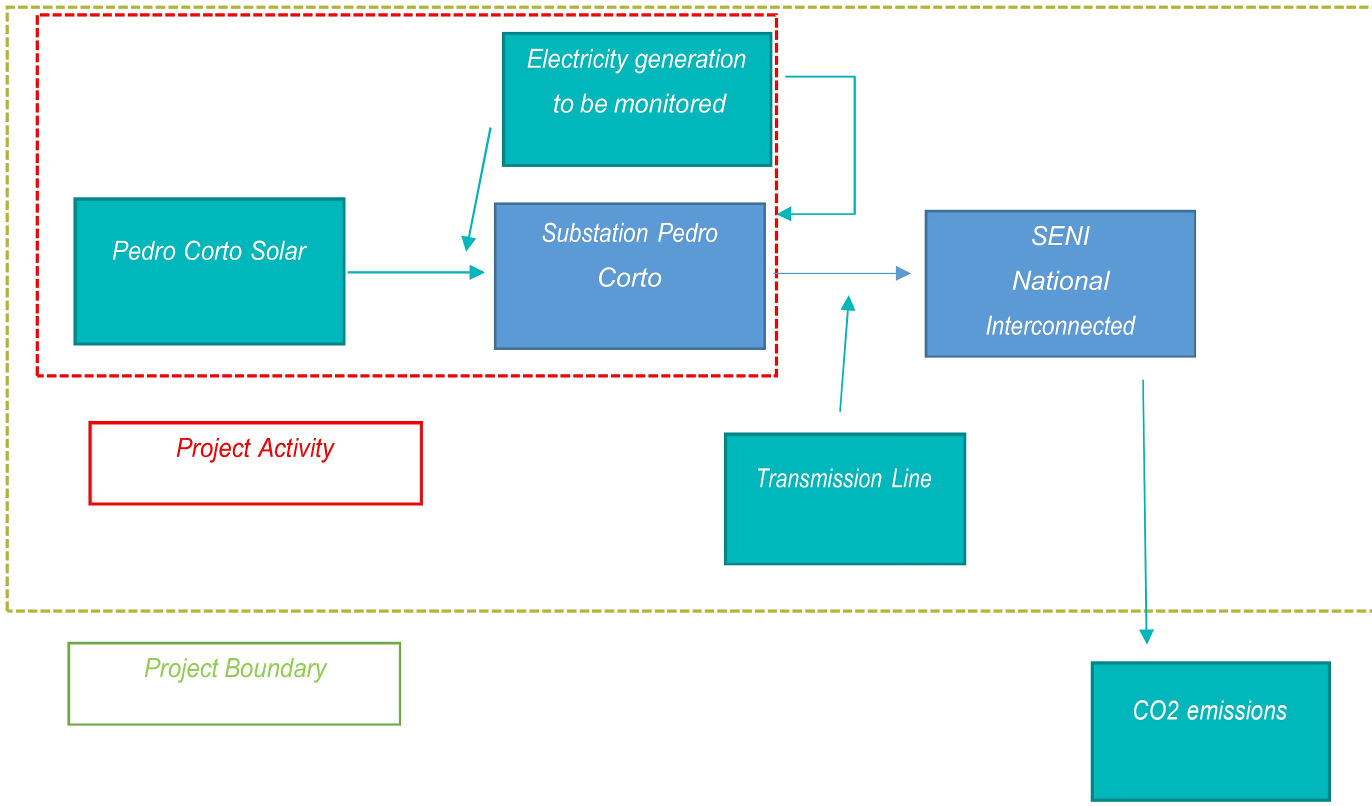
According to the methodology ACM0002 (Version 21.0), the spatial extent of the project boundary includes the project power plant/unit and all power plants/units connected physically to the electricity system that the proposed project power plant is connected to.

Additionally, according to the “Tool to calculate the emission factor for an electricity system” (Version 7.0, Tool 07), a grid/project electricity system is defined by the spatial extent of the power plants that are physically connected through transmission and distribution lines to the project activity and that can be dispatched without significant transmission constraints.

In this case, the project activity is connected to the National Interconnected Electrical System (SENI). Thus, Pedro Corto Solar site is connected to the SENI and included in the project boundary.

The following diagram shows the project boundary based on the description above:



B.4. Establishment and description of baseline scenario

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As stated in the approved methodology ACM0002 "Grid-connected electricity generation from renewable sources", version 21.0.0: If the project activity is the installation of a Greenfield power plant with or without a BESS as described under paragraph 4(a) or paragraph 5(a), the baseline scenario is electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in TOOL07 ,.(Version 7.0, Tool 07)

Therefore, the baseline scenario consists of the electricity that would have been generated and delivered to the grid in the absence of the proposed project activity by:

1) Other plants currently connected to the SENI; and

2) New capacity additions to the SENI.

Hence, the baseline scenario is identified as the continuation of the common practice of power generation, i.e., mainly thermal power plants (Fuel oil, natural gas and coal) that emit large quantities of carbon dioxide (CO_2) to the atmosphere.

The baseline based on the Tool for the demonstration and assessment of additionality (Version 7.0, Tool 01) and the Combined Tool to Identify the Baseline Scenario and Demonstrate Additionality (Version 7.0, Tool 02), would be the electricity generated and delivered to the grid by the other plants currently connected to the grid which are mainly fuel oil, natural gas and coal. Find attached the list of plants connected to the Dominican electric grid in OC-SENI annual report.

B.5. Demonstration of additionality

B.5.1 Prior Consideration

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Pedro Corto Solar is a regular project and therefore exempt from any kind of prior consideration of revenues from Gold Standard certification checks. In order to have a clear visualization of the project, the following tables show project events, Gold Standard events and permits through time.

The following table shows the history of events for Pedro Corto Solar to show the project events and Gold Standard events.

Date	Key Event	Hiring Firm	Executing Firm	Reference Document Sharepoint
January 17, 2019	Land Acquisitions- Land Owners (Land negotiation and contracts)	Irradiasol Dominicana SRL	Soventix Caribbean SRL	Contracts
February 20, 2019	Creation of Irradiasol Dominicana SRL	Irradiasol Dominicana SRL	Soventix Caribbean SRL	Commercial Register
March 8, 2019	Incorporation of Irradiasol Dominicana SRL	Irradiasol Dominicana SRL	Soventix Caribbean SRL	Commercial Register
August 15, 2019	Request Provisional Concession*	Irradiasol Dominicana SRL	Soventix Caribbean SRL	Request Provisional Concession Receipt
September 15, 2019	Start of the preparation of the Basic	Irradiasol Dominicana SRL	Soventix Caribbean SRL	Descriptive Memory

	Engineering of the Project			
October 22, 2019	Investment Decision Date	Irradiasol Dominicana SRL	Canadian Solar Spain SL and Soventix Caribbean SRL	Executed SHA Irradiasol CS SVC English
January 13,2020	Request for the Terms of Reference for the EIA (Environmental Impact Study) and contract signed with EMPACA to perform the EIA.	Irradiasol Dominicana SRL	EMPACA	Quotation
February 18, 2020	Procurement of Provisional Concession	Irradiasol Dominicana SRL	Soventix Caribbean SRL	Procurement of Provisional Concession
May 13, 2020	Request for Letter of Non-objection of Interconnection	Irradiasol Dominicana SRL	Soventix Caribbean SRL	Non objection + Technical Report
June 18, 2020	Procurement of Land Use Permit given by local Council	Irradiasol Dominicana SRL	Soventix Caribbean SRL	Non objection of land use by local Council
July 6, 2020	Request for the Environmental License	Irradiasol Dominicana SRL	Soventix Caribbean SRL	Request for the Environmental License receipt
September 30, 2020	Start of Environmental Impact Study (EIA)	Irradiasol Dominicana SRL	EMPACA	TdR

October 19, 2020	Procurement of Letter of Non-objection of Interconnection	Irradiasol Dominicana SRL	Soventix Caribbean SRL	Non objection + technical report
October 20, 2020	Decision to certify the Project as Gold Standard	Irradiasol Dominicana SRL	Soventix Caribbean SRL	Project Ideas Notes Pedro Corto
December 1, 2020	First Stakeholder´s Meeting in Pedro Corto	Irradiasol Dominicana SRL	Soventix Caribbean SRL	Pedro Corto Stakeholder Invitation English
July 14, 2021	First Preliminary Review with SustainCert	Irradiasol Dominicana SRL	Soventix Caribbean SRL	Preliminary Review Pedro Corto
November 24, 2021	Project appears as listed in the Gold Standard Impact Registry	Irradiasol Dominicana SRL	Gold Standard	Listing date GS
December 1, 2021	Start of Financial Evaluation of the Project	Irradiasol Dominicana SRL	Soventix Caribbean SRL	Financial resume
December 8, 2021	Request for Definitive Concession	Irradiasol Dominicana SRL	Soventix Caribbean SRL	Letter request for definitive concession
December 17, 2021	Procurement of Environmental License (Environmental license granted)	Irradiasol Dominicana SRL	Soventix Caribbean SRL	Environmental License
February 8, 2022	Validation starts/First Meeting	Soventix Caribbean SRL	Epic Sustainability	-

March 28, 2022	Meeting Stakeholders for Validation	Soventix Caribbean SRL	Epic Sustainability	NGO Questions

*The Investment Decision and Date is below because the negotiations to sign the contract between the partners took longer than expected. Even though the contract had not yet been signed the course of the project was not stopped, the Provisional Concession and preparation of the Engineering was done.

The following table shows the different documents and mandatory requirements in order to operate the project, starting from the conception of the idea of the project.

Legal Project Performance Pedro Corto Solar

Documents Required	Description	Status
Provisional Concession Request to the CNE (Comisión Nacional de Energía)	Permission to perform studies	Completed
Land Use Permit given by local Council Ayuntamiento del Distrito Municipal Pedro Corto	Permission of the local authority for the use of the land	Completed
Land Acquisitions-Land Owners	Land negotiation and contracts	Completed
Environmental License (Ministerio de Medio Ambiente y Recursos Naturales)	Permission to construct Project	Completed
Definitive Concession by the Executive Power via CNE (Comisión Nacional de Energía)	Permission that grants the operation of the Project through the Concession period	In process
Construction Permit by the MOPC (Ministerio de Obras Públicas y Comunicaciones)	Permission to construct the project, validating buildings, lay outs, etc.	Not started, to be obtained after the Definitive Concession
Commercial Operation Permit by the SIE (Superintendecia de Electricidad)	Process through which the technical aspects of the power plant and metering system are validated by the Coordination Body of the Electrical Sector (OC Organismo Coordinador	Not started, to be obtained once the project is built and ready for Interconnection

	del Sistema Eléctrico Nacional Interconectado de la República Dominicana) and the SIE (Superintendecia de Electricidad). The plant is granted with permanent interconnection to the grid.	

B.5.2 Ongoing Financial Need

>>

The additionality of the project activity is demonstrated and assessed applying the CDM "Tool for the demonstration and assessment of additionality" (Version 7.0.0, Tool 01), as stated in the methodology ACM0002 (version 21.0.0) and accepted by Gold Standard.

The tool provides a stepwise approach to demonstrate and assess additionality:

Step 0. Demonstration whether the proposed project activity is the first-of-its-kind;

Step 1. Identification of alternatives to the project activity;

Step 2. Investment analysis to determine that the proposed project activity is either:

(1) Not the most economically or financially attractive, or

(2) Not economically or financially feasible;

Step 3. Barriers analysis; and

Step 4. Common practice analysis.

Step 0: Demonstration whether the proposed project activity is the first-of-its-kind

This step is optional. It is not applied for this project activity and therefore it is considered that the proposed project activity is not the first-of-its-kind.

Step 1: Identification of alternatives to the project activity consistent with current laws and regulations

Realistic and credible alternatives to the project activity(ies) are defined through the following sub-steps:

Sub-step 1a: Define alternatives to the project activity

According to the Tool for the demonstration and assessment of additionality, project activities applying this tool in context of approved methodology ACM0002, only need to identify that there is at least one credible and feasible alternative that would be more attractive than the proposed project activity.

Result of Step 1a: For the project proponent, the possible alternatives to the proposed project include:

Alternative 1: The proposed project activity undertaken without being registered as a GS project activity.

Alternative 2: Continuation of the current situation. In this particular case, the project activity will not be constructed, and the power will be exclusively supplied by the operation of the Sistema Eléctrico Nacional Interconectado (SENI) connected power plants and by the addition of new power plants.

Alternative 3: Other realistic and credible alternative scenario(s) to the proposed CDM project activity scenario that deliver outputs services or services with comparable quality, properties and application areas, taking into account, where relevant, examples of scenarios identified in the underlying methodology. In this particular case, other credible alternative scenarios would be hydro or wind projects. As per hydro projects, the zone where Pedro Corto Solar will be implemented has no feasible water resources, also the government can only give hydroelectric concessions to the private sector to plants that are below 5 MW. As per wind projects, the zone where Pedro Corto Solar will be implemented has a very low wind potential as can be seen from the figure below.

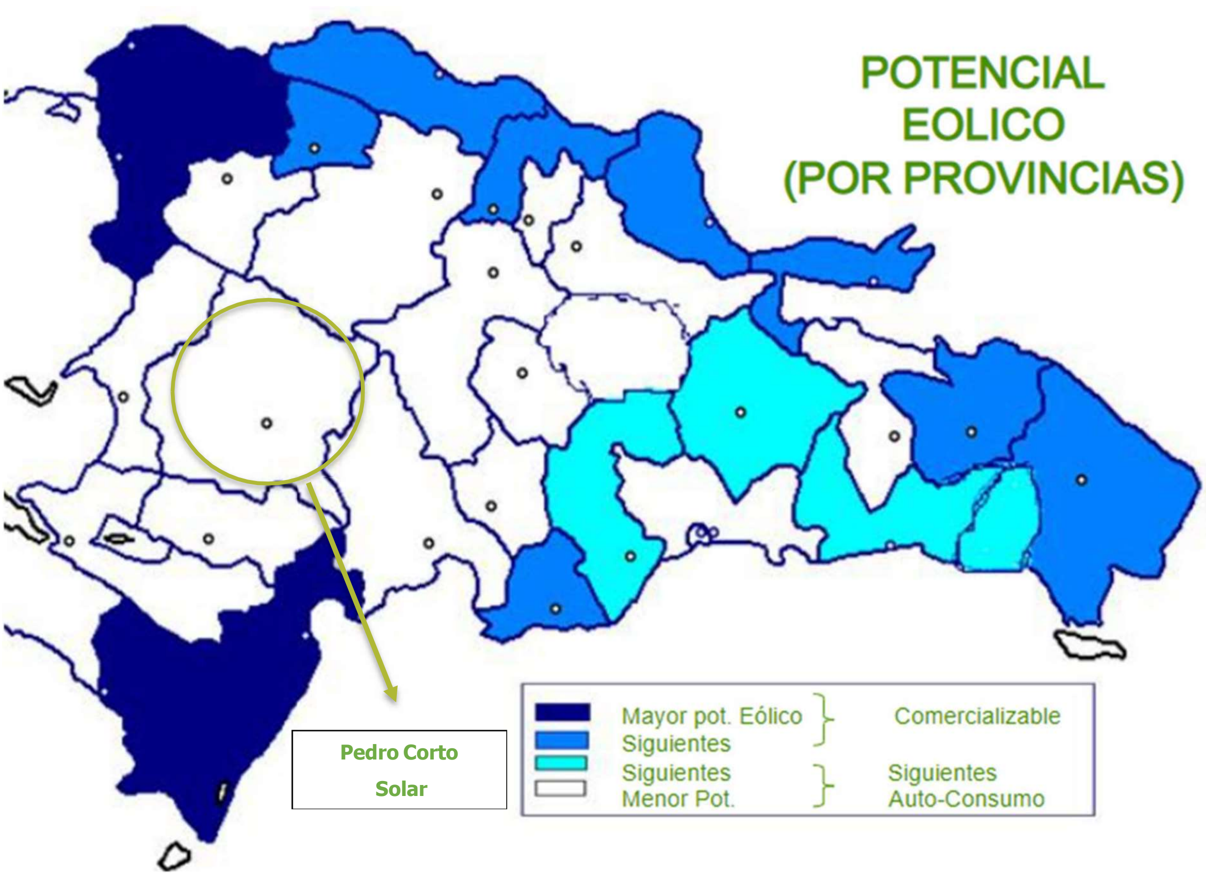

Figure 6. Wind potential by provinces (Source Comisión Nacional de Energía CNE)

<u>Sub-step 1b</u>: Consistency with mandatory laws and regulations

The implementation and operation of the project activity affects two different groups of power plants in the interconnected national system as represented in the baseline scenario by the combined margin:

1. Existing power plants whose current electricity generation would be affected by the proposed project activity; and
2. Prospective power plants whose construction and future operation would be affected by the proposed project activity.

Both alternatives, i.e., the proposed project activity undertaken without being registered as a Gold Standard project and the continuation of the current situation, are in compliance with all of the mandatory laws and regulations, this is supervised by the corresponding authorities in the Dominican Republic:

1. Ministry of Environment and Natural Resources of Dominican Republic: Compliance with all environmental laws and standards. The project proponent is is beginning the process of obtaining the environmental license through the Environmental Impact Study.
2. Superintendencia de Electricidad - SIE: The project proponent will receive the connection approval once the project is ready to begin commercial operation.
3. Comisión Nacional de Energía – CNE- and the SIE: The CNE responsible for designing and implementing the energy policy of the state, and the SIE is responsible for overseeing the operation of the electricity market and setting rates to the final consumer. They are also responsible for issuing concessions, which the project proponent is in the process of obtaining.

The proposed project is fully in compliance with the current laws and regulations of the country.

<u>Result of Step 1b</u>: Both alternatives identified in sub-step 1a are in compliance with mandatory laws and regulations:

<u>Alternative 1</u>: The proposed project activity undertaken without being registered as a GS project activity. The project is consistent with the current laws and regulations of the Dominican Republic. Concerning the General Electricity Law, the project activity complies with all applicable legal and regulatory requirements. The Law on Incentives for the Development of Renewable Energy Sources and Special Regimes, can be applied to the proposed project and therefore incentives will be sought once it is required. Thus, Alternative 1 is a viable option and is included for further analysis.

<u>Alternative 2</u>: Continuation of the current situation: In this case, the project activity will not be constructed, and the power will be supplied exclusively by the operation of the country, Sistema Eléctrico Nacional Interconectado (SENI) connected power plants and by the addition of new power plants.

Then, electricity not generated by the project activity would be dispatched to the SENI by existing plants (operating margin) and by future plants (build margin). As operations in existing plants and construction of new plants are valid alternatives under applicable laws and regulations, the alternative scenario is in compliance with the regulatory framework for electricity supply by existing or future SENI plants. Thus, Alternative 2 is a viable option and is included for further analysis.

Step 2: Investment analysis

To conduct the investment analysis, the following Sub-steps are used:

Sub-step 2a: Determine appropriate analysis method

According to CDM "Tool for the demonstration and assessment of additionality" (Version 7.0.0, Tool 01), three options can be applied for the investment analysis: the simple cost analysis, the investment comparison analysis and the benchmark analysis.

The project activity generates financial and economic benefits other than VER revenues, therefore the simple cost analysis (Option I) is not applicable. The corresponding CDM Methodological tool: Investment analysis (Version 12.0, Tool 27), states that: "if the alternative to the project activity is the supply of electricity from a grid this is not to be considered an investment and the benchmark approach is considered appropriate". Thus, Option II (investment comparison analysis) is also not applicable, and the benchmark analysis (Option III) is chosen to prove additionality.

Sub-step 2b – Option III: Apply benchmark analysis

The additionality tool requires an identification of the most appropriate financial indicator. For the case of a power plant that would supply energy to the grid, the most appropriate indicator is the internal rate of return (IRR) as it characterizes the rate of return on invested capital. In this analysis an equity IRR is calculated in accordance with the additionality tool and the corresponding methodological tool as indicated above. Taxation is included as an expense in the IRR calculation, for example, the IRR is determined as a post-tax indicator.

In accordance with the "Methodological tool: Investment analysis" (Version 12.0, Tool 27) a default value for the expected return on equity is used for the benchmark. The relevant benchmark for energy projects in Dominican Republic (Group 1 as given in the methodological tool) is 11.73% in real terms. As per the methodological tool, since the investment analysis is carried out in nominal terms, the real term values provided can be converted to nominal values by adding the inflation rate. The inflation target rate of the Central Bank of Dominican Republic – CBDR, for the duration of the crediting period is available; then the average forecasted inflation rate of 4.00% was calculated for the next five years, based on the forecasts for the period from 2020 to 2025.

The benchmark, i.e., the Nominal Return on Equity, is therefore given as 15.73%.

Sub-step 2c: Calculation and comparison of financial indicators

For the financial analysis the main cash outflows are given by the investment, the ongoing O&M costs and taxes. The cash inflows are generated from the revenues of electricity sales (which depend on power generation and electricity prices).

Input values for the investment analysis

The financial structure is applied as suggested by the "Methodological tool for Investment analysis" (version 12.0, Tool 27). In this case the debt of the project is equal to 30%. Therefore, the terms of debt structure are given in the following table.

Parameter	Value	Source
Total Investment (USD)	$67,466,630	Investment decision
% Debt	70%	
Equity Amount (USD)	$20,239,989	Calculated in nominal terms
Debt Amount (USD)	$47,226,641	Calculated in nominal terms
Principal (Annual)	$2,023,998.9	Annual principal during 10 years

The table below lists the parameters and values used for carrying out the investment analysis. All sources and further details are provided in the Investment Analysis spreadsheets.

An operational period of 25 years is used for the financial analysis, given by a Draft Engineering, Procurement and Construction Agreement Proposal – Draft EPC Proposal.

At the end of the project lifetime, no fair value is included since the equipment will be devaluated.

Electricity Generation		
Capacity	82.5	MW
Net Energy Generation	151.2	GW h/year
Tariffs		
Electricity Sales		
Energy Price (average)	$75	USD/MWh
Capacity Price (average)	0	USD/kW-month
Renewable Energy Certificates	0	USD/MWh
Investment		
Total Capital Costs		
Total Investment (including EPC contractand project rights)	$67,466,630	USD
Equipment cost (for depreciation only)	$53,973,304	USD
Operation Costs and Expenses		
Operation Costs		
O&M (Substation and WTG)	$1,070,850	USD/year
Insurance	$674,666.3	USD/year
Land Costs	1.77%	% of income
Taxes		
Income Taxes	27	%
Depreciation		
Annual depreciation Civil Works + Infrastructure	20	Years
Annual depreciation equipment	20	Years
Financial Parameters		
Macro		
Inflation rate (forecast BCRD)	4.00	%
Benchmark		
Return on Equity (real terms)	11.73	%
Inflation Adjustment	4.00	%
BENCHMARK: Nominal Return on Equity	15.73	%

Result of the investment analysis

Based on the parameters above, the Internal Rate of Return (post-tax equity IRR) is calculated as 11.55%, which is below the benchmark rate of 15.73%.

Sub-step 2d: Sensitivity analysis A sensitivity analysis is carried out by varying the following key parameters to analyze the impact on the equity IRR:

Energy generation (MWh): The quantity of energy generated is varied.

Energy tariff (USD/MWh): The energy tariff is varied.

Investment costs (USD): the complete investment cost is varied.

O&M costs (USD/year): the complete operational cost is varied.

The following table shows a sensitive analysis for the resulting IRR by varying the main parameters +/- 10%.

Variation of Electricity Generation	Base	+10%
IRR	11.55%	15.12%

Variation of Electricity Tariff	Base	+10%
IRR	11.55%	14.84%

Variation of Investment Costs	Base	-10%
IRR	11.55%	15.05%

Variation of O&M Costs	Base	-10%
IRR	11.55%	11.91%

As can be seen, the benchmark is not reached in these scenarios; thus, it can be concluded that the project activity is not financially attractive.

Step 3: Barrier analysis

The project activity does not apply a barrier analysis.

Step 4: Common practice analysis has been done with the“Methodological Tool Common Practice” (Version 03.1, Tool 24). The applicable geographical area for the assessment of common practice analysis corresponds to thehost country Dominican Republic.

According to the guidelines for Common practice analysis, the following steps are applied:

Step 1: Calculate applicable capacity or output range as +/-50% of the total design capacity or output of the proposed project activity.

For the common practice analysis, the installed capacity of 82.5 MW is used, and therefore the applicable range as +/-50% of the capacity is 41.25 MW to 123.75 MW.

Step 2: Identify similar projects (both CDM and non-CDM).

The information about all operational power plants connected to the SENI is obtained from Organismo Coordinador del Sistema Eléctrico Nacional, the document was accessed on January 2021. The following power plants have a power generation capacity within the capacity range identified in Step 1:

Power plants in the same capacity range (+/-50%) (41.25-123.75 MW)
SENI-Installed Capacity per unit (MW)-2020

Number	Power Plant	Date of Commissioning	Source	Installed Capacity MW	Registered	Different
1	CEPP 2	1994	Fuel Oil #6	58.1	No	Yes
2	CESPM 1	2021	Natural Gas/Fuel Oil #2	100	No	Yes
3	CESPM 2	2021	Natural Gas/Fuel Oil #2	100	No	Yes
4	CESPM 3	2021	Natural Gas/Fuel Oil #2	100	No	Yes
5	METALDOM	1992	Fuel Oil #6	42	No	Yes

6	LOS MINA 5	1997	Natural Gas	118	No	Yes
7	LOS MINA 6	1997	Natural Gas	118	No	Yes
8	LOS MINA 7	2017	Natural Gas	123.25	No	Yes
9	BARAHONA CARBÓN	2001	Coal	51.9	No	Yes
10	LARIMAR	2016	Wind	49.5	Yes	Yes
11	HAINA TG	2000 (Rehabilitation)	Fuel Oil #2	100	No	Yes
12	SULTANA DEL ESTE	2001 (Rehabilitation)	Fuel Oil #6	68.4	No	Yes
13	LOS COCOS 2	2012	Wind	52	Yes	Yes
14	LARIMAR 2	2019	Wind	48.3	Yes	Yes
15	JIGUEY 1	1992	Water	49	No	Yes
16	JIGUEY 2	1992	Water	49	No	Yes
17	TAVERA 1	1988	Water	48	No	Yes
18	TAVERA 2	1988	Water	48	No	Yes
19	LA VEGA	2001	Fuel Oil #6	92.1	No	Yes
20	PALAMARA	2000	Fuel Oil #6	107	No	Yes
21	AGUA CLARA	2019	Wind	52.5	No	Yes
22	PIMENTEL 3	2011	Natural Gas/Fuel Oil #6	51.2	No	Yes
23	MONTE RÍO	2003	Fuel Oil #6	101.5	No	Yes
24	LOS ORÍGENES	2013	Natural Gas/Fuel Oil #6	60.7	No	Yes
25	MONTECRISTI SOLAR	2018	Sun	58	No	No
26	GUANILLO	2019	Wind	52.5	Yes	Yes
27	PARQUE EÓLICO LOS GUZMANCITOS	2020	Wind	48	No	Yes
28	PARQUE FV MATA DE PALMA	2020	Sun	66.9	No	No
29	ESTRELLA DE MAR 2	2012	Natural Gas/Fuel Oil #6	111.3	No	Yes

At the end of Step 2, the number of projects similar are 2: Montecristi Solar and Mata de Palma FV. Larimar, Larimar 2, Los Cocos 2 and Guanillo are registered wind projects.

Step 3: Within the projects identified in Step 2, identify those that are neither registered CDM project activities, project activities submitted for registration, nor project activities undergoing validation. Note their number N_{all}.

From the total of 29 power plants in the same capacity range (Step 2), there are 4 projects submitted for registration or undergoing validation; thus N_{all} = 25. (for example, in the above table, all projects that are not greyed out and as can be seen in the table below)

Project	Registration	ID Number	Status
Larimar Wind Farm Project (Larimar and Larimar 2)	VCS (VERRA)	ID 1644	Registered
Los Cocos Wind Farm Project	CDM	ID 7093	Registered
Los Cocos II Wind Farm Project	CDM	ID 7100	Registered
El Guanillo	CDM	ID 0175	Registered

Step 4: Within plants identified in Step 3, identify those that apply technologies that are different to the technology applied in the proposed project activity. Note their number as N_{diff}.

As per the guidelines, different technologies are technologies that deliver the same output and differ by at least one of the following (as appropriate in the context of the measure applied in the proposed CDM project and applicable geographical area):

1. Energy source/fuel (example: energy generation by different energy sources such as wind and hydro and different types of fuels such as biomass and natural gas);
2. Feed stock (example: production of fuel ethanol from different feed stocks such as sugar cane and starch, production of cement with varying percentage of alternative fuels or less carbon-intensive fuels);
3. Size of installation (power capacity) / energy savings:
   - Micro
   - Small

- Large.

4. Investment climate in the date of the investment decision, inter alia:

   - Access to technology;
   - Subsidies or other financial flows;
   - Promotional policies;
   - Legal regulations;

5. Other features, inter alia:

   - Nature of the investment (example: unit cost of capacity or output is considered different if the costs differ by at least 20 %).

An important distinction can be made by the "Energy source/fuel", since there are other technologies involved, using other resources (i.e., thermal power plants, hydro power plants or wind). Thus, from the 25 power plants identified above (N_{all}), all but 2 power plants can be excluded (for example, the two solar power plants highlighted in yellow in the above table cannot be excluded since they have the same Energy source/fuel), therefore N_{diff} = 23.

Step 4: Calculate factor $F=1-N_{diff}/N_{all}$ representing the share of plants using technology similar to the technology used in the proposed project activity in all plants that deliver the same output or capacity as the proposed project activity.

As per the guidelines, a project is considered a common practice within a sector in the applicable geographical area if the factor F is greater than 0.2 and N_{all} - N_{diff} is greater than 3. Since N_{all} = 25 and N_{diff} = 23,

$N_{all} - N_{diff} = 2$ (and $F=0.08 < 0.2$)

Based on the second criterion, the project is not a common practice.

Conclusion of the additionality analysis

Since the project activity is not financially attractive (Step 2) and the common practice analysis shows that it is not business-as-usual (Step 4), the proposed project activity is **additional.**

B.6. Sustainable Development Goals (SDG) outcomes

Relevant Target/Indicator for each of the four SDGs

The following relevant SDG contributions were identified and will be addressed by the project. The three SDGs that were identified for the project are SDG 7, SDG 8, SDG 9 and SDG 13.

SDG 7: Affordable and Clean Energy

Relevant Target	Indicator
7.2: By 2030, increase substantially the share of renewable energy in the global energy mix.	Total renewable energy amount delivered to the grid.

For SDG7 Affordable and Clean Energy relevant target 7.2 was chosen. This relevant target states that "By 2030, increase substantially the share of renewable energy in the global energy mix", therefore the project will contribute to the total renewable energy amount delivered to the grid and will increase renewable energy sources in the energy mix from the Dominican Republic. As a conclusion, there is a direct connection between the increase of renewable energy share and the total amount of solar energy delivered to the grid.

SDG 8: Decent Work and Economic Growth

Relevant Target	Indicator
8.5: By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities and equal	Total number of jobs created during the construction and operation and maintenance phase.

pay for work of equal value.	

For SDG 8 Decent Work and Economic Growth target 8.5 was chosen. Therefore, the project will contribute to achieve full and productive employment and decent work for all men and women, including young people and persons with disabilities. The project will contribute with jobs for the local people during the construction and operation and maintenance phase, equal opportunities will be available without discrimination on any basis.

SDG 9: Industries, Innovation and Infrastructure

Relevant Target	Indicator
9.4: By 2030, upgrade infrastructure and retrofit industries to make them	Total number of renewable energy plants in the Dominican energy mix.
sustainable, with increased resource-use efficiency and greater adoption of clean and environmentally sound technologies and industrial processes, with all countries taking action in accordance with their respective capabilities	

There is a connection between the number of renewable energy plants in the energy mix of the Dominican Republic and the energy that would have been generated by non-renewable resources like fossil fuel and gas. Therefore, the project will contribute to the total renewable energy amount delivered to the grid and the increase of renewable energy share to the energy mix of the country.

Note: It is relevant as this project will contribute to the energy mix with renewable energy that industries will use to make industrial processes sustainable. If more renewable energy plants are introduced less non-renewable resources like fossil fuels and gas will be used. As more renewable energy plants are introduced into the Dominican energy mix, industries and processes will use this clean energy to produce and therefore industrial processes will become more sustainable.

SDG 13: Climate Action

Relevant Target	Indicator
13.2: Integrate climate change measures into national policies, strategies and Planning	Amount of CO_2 emissions reduced by the project per year.

B.6.1 Explanation of methodological choices/approaches for estimating the SDG Impact

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SDG 7: Affordable and clean energy

Indicator: 7.2.1 Renewable energy share in the total final energy consumption.

In the baseline scenario the electricity would have been generated by the national grid which is mainly fed with electricity generated through fossil-fuel sources. The project will annually produce and supply clean, renewable and sustainable energy to the national electric grid.

Applied Methodology/Approach	Equation/Calculation
Monthly reports from the Organismo Coordinador del Sistema Eléctrico of the Dominican Republic	$EG_{facility,y}$ (MWh/y) = Quantity of net electricity generation supplied by the project plant to the grid in year y

SDG 8: Decent Work and Economic Growth

Indicator: 8.5 By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities and equal pay for work of equal value

In the baseline scenario no such employment opportunities would have been available for the local communities. The project will benefit the local community as residents will be given priority for employment opportunities, based on their qualifications and skills.

Applied Methodology/Approach	Equation/Calculation
Contracts made in accordance with the principles of gender equality and transparency established by Dominican Republic law	Number of local people employed

SDG 9: Industries, Innovation and Infrastructure

Indicator: 9.4 By 2030, upgrade infrastructure and retrofit industries to make them sustainable, with increased resource-use efficiency and greater adoption of clean and environmentally sound technologies and industrial processes, with all countries taking action in accordance with their respective capabilities

In the baseline scenario the electricity used by industries in the country would have been generated by the national grid which is mainly fed with electricity generated through fossil-fuel sources. The project and other renewable energy projects added to the grid annually will produce and supply clean, renewable and sustainable energy to the national electric grid and therefore industries will benefit from this type of energy.

Applied Methodology/Approach	Equation/Calculation
Annual reports from the Organismo Coordinador del Sistema Eléctrico Nacional of the Dominican Republic	1. Total number of renewable energy projects added the Dominican grid

SDG 13: Climate Action

Applied Methodology/Approach	Equation/Calculation
ACM0002: Grid-connected Electricity Generation from Renewable Sources Version 20.0	As described below in the document.

The methodology ACM0002 assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. Thus, baseline emissions include only CO_2 emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity.

Baseline emissions are calculated as follows:

$$\mathbf{BE_y = EG_{PJ,y} \times EF_{grid,\,CM,\,y}}$$

Where:

BE_y = Baseline emissions in year y (tCO_2)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y (MWh)

$EF_{grid, CM, y}$ = Combined margin CO2 emission factor for grid connected power generation in year y (tCO2/MWh)

According to the methodology, if the project activity is the installation of a Greenfield power plant, the quantity of net electricity generation is determined as follows:

$$\mathbf{EG_{PJ,y} = EG_{facility,y}}$$

Where:

$EG_{PJ, y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EG_{facility, y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

Additionally, the methodology states that the grid emission factor should be calculated using the latest version of the "Tool to calculate the emission factor for an electricity system". Thus, the grid emission factor is calculated as a combined margin emission factor using Version 7.0 of the Tool 07 and according to the following steps:

Project participants shall apply the following six steps:

Step 1: Identify the relevant electricity systems;

Step 2: Choose whether to include off-grid power plants in the project electricity system (optional);

Step 3: Select a method to determine the operating margin (OM);

Step 4: Calculate the operating margin emission factor according to the selected method;

Step 5: Calculate the build margin (BM) emission factor;

Step 6: Calculate the combined margin (CM) emission factor.

<u>Step 1</u>: Identify the relevant electricity systems

For determining the electricity emission factors, a project electricity system is defined by the spatial extent of the power plants that are physically connected through transmission and distribution lines to the project activity (Pedro Corto Solar project) and that can be dispatched without significant transmission constraints. In this case, the project electricity system is the Organismo Coordinador del Sistema Eléctrico Nacional OC-SENI.

For the purpose of determining the OM emission factor, the CO_2 emission factors for net electricity imports from the connected electricity systems are assumed as 0 tCO_2/MWh, according to option (a) given by the Tool. Nevertheless, the SENI grid does not have any imports from other electric grids.

On the other hand, and following the before mentioned tool, electricity exports should not be subtracted from electricity generation data used for calculating and monitoring the baseline emission factors. Nevertheless, the SENI grid does not have any exports to other electric grids.

Step 2: Choose whether to include off-grid power plants in the project electricity system

Either of the following two options can be used to calculate the OM and the BM emission factor:

a. Option I: Only grid power plants are included in the calculation.

b. Option II: Both grid power plants and off-grid power plants are included in the calculation.

In accordance with the tool, this step is optional. For the proposed project activity, off-grid power plants are not included in the project electricity system. In this case, Option I is chosen.

Step 3: Select a method to determine the OM

In accordance with the tool, the calculation of the operating margin emission factor ($EF_{grid,OM,y}$) is based on one of the following methods:

a. Simple OM

b. Simple Adjusted OM

c. Dispatch Data Analysis OM

d. Average OM

For the project activity, the simple OM method is applied, using the ex-ante data vintage:

Ex ante option: if the ex-ante option is chosen, the emission factor is determined once at the validation stage, thus no monitoring and recalculation of the emissions factor during the crediting period is required. For grid power plants, use a 3-year generation-weighted average, based on the most recent data available at the time of submission of the CDM-PDD to the DOE for validation. For off-grid power plants, use a single calendar year within the five most recent calendar years prior to the time of submission of the CDM-PDD for validation.

The Tool states that the Simple OM method can only be used where low-cost/must run resources constitute less than 50% of total grid generation (excluding electricity generated by off-grid power plants) in: 1) average of the five most recent years, or 2) based on long-term averages for hydroelectricity production.

All power plants connected to the SENI are included. Power plants registered as CDM project activities are also included as suggested by the tool. Historical data of the three most recent years is available at the Coordinating Organization OC-SENI web site (www.oc.org.do).

Step 4: Calculate the OM emission factor according to the selected method

The simple OM method can only be used if any one of the following requirements is satisfied:

Low-cost/must-run resources constitute less than 50 per cent of total grid generation (excluding electricity generated by off-grid power plants) in the five most recent years. LCMR (low cost-must run) plants are defined as power plants with low marginal generation costs or dispatched independently of the daily or seasonal load of the grid. In this case, included hydro, wind, solar and low-cost biomass for the Dominican Republic.

$$EF_{grid,\ OM\text{-}simple,y} = \frac{\sum EG_{m,y} \cdot EF_{EL,m,ym}}{\sum EG_{m,ym}}$$

Where:

$EF_{gridOM\text{-}simple,y}$ = Simple operating margin CO_2 emission factor in year y (t CO_2/MWh)

$EG_{m,y}$ = Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)

$EF_{EL,m,y}$ = CO_2 emission factor of power unit m in year y (tCO_2/MWh)

M = All grid power units serving the grid in year y except low-cost/must-run power units

Y = The relevant year as per the data vintage chosen in step 3.

As per the document ASB0047-2020 "Standarized baseline Grid Emission Factor for the Dominican Republic" (V 01.0) provides the values of the CO_2 emission factors for the National Interconnected Electrical System (SENI) of the Dominican Republic.

The scope of this standardized baseline covers the grid emission factor for the SENI. It was derived using the ex-ante data vintage option of the "TOOL07: Tool to calculate the emission factor for an electricity system", version 7.0 based on 2014 – 2016 data vintage. Clean development mechanism (CDM) project activities and programs of activities can apply this standardized baseline under the following conditions:

a. The project activity is implemented in the Dominican Republic and is connected to the SENI;

b. The CDM approved methodology that is applied to the project activity requires the determination of CO_2 emission factor(s) through the application of the grid tool;

c. The project activity uses the ex-ante options for both the operating margin and build margin grid emissions factors, as described in the grid tool, and therefore no monitoring or recalculation of the emission factor during the crediting period is required.

Pedro Corto Solar complies with all three conditions and therefore the standardized values provided by this document are used.

The latest approved and valid values of this standardized baseline are the only values of the CO_2 emission factor(s) that shall be applied for the SENI.

Parameter	**Unit**	**Description**	**Applicable project types**	**Applicable values (First, Second and Third Crediting Period)**
$EF_{grid, OM, y}$	tCO_2/MWh	Operating margin CO_2 emission factor for the SENI	All project activities	0.6300

Step 5: Calculate the BM emission factor

In terms of vintage of data, option 1 of the tool is chosen, i.e., the ex-ante approach:

Option 1: For the first crediting period, calculate the build margin emission factor ex ante based on the most recent information available on units already built for sample group m at the time of CDM-PDD submission for validation. For the second crediting period, the build margin emission factor should be updated based on the most recent information available on units already built at the time of submission of the request for renewal of the crediting period to the DOE. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used. This option does not require monitoring the emission factor during the crediting period.

Capacity additions from retrofits of power plants are not included in the calculation of the Build Margin - BM emission factor.

The sample group of power units m used to calculate the BM is determined as per the following procedure, consistent with the data vintage selected above:

a) Identify the set of five power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently ($SET_{5\text{-}units}$) and determine their annual electricity generation ($AEGSET_{\text{-}5\text{-}units}$, in MWh).

b) Determine the annual electricity generation of the project electricity system, excluding power units registered as CDM project activities (AEG_{total}, in MWh). Identify the set of power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently and that comprise 20% of AEG_{total} (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation) ($SET_{\geq 20\%}$) and determine their annual electricity generation ($AEG_{SET\text{-}\geq 20\%}$, in MWh).

c) From SET5-units and SET≥20% select the set of power units that comprises the larger annual electricity generation (SET_{sample}).

   Identify the date when the power units in SET_{sample} started to supply electricity to the grid. If none of the power units in SET_{sample} started to supply electricity to the grid more than 10 years ago, then use SET_{sample} to calculate the BM. In this case, ignore steps (d), (e) and (f).

The calculation is made as follows:

$$EF_{grid,BM,y} = \frac{\sum_m EG_{m,y} \times EF_{EL,m,y}}{\sum_m EG_{m,y}}$$

Where:

$EF_{grid,BM,y}$ = BM CO_2 emission factor in year y (tCO_2/MWh)

$EG_{m,y}$ = Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)

$EF_{EL,m,y}$ = CO_2 emission factor of power unit m in year y (tCO_2/MWh)

M = Power units included in the BM

Y = Most recent historical year for which electricity generation data is available

Parameter	Unit	Description	Applicable project types	Applicable values (First, Second and Third
				Crediting Period)
$EF_{grid, BM, y}$	tCO_2/MWh	Build margin CO_2 emission factor for the SENI	All project activities	0.5962

<u>Step 6:</u> Calculate the CM emission factor

According to the Tool, the calculation of the Combined Margin - CM emission factor is based on one of the following methods:

(a) Weighted average CM

(b) Simplified CM

In this case, Option (a) is chosen. Thus, the CM emission factor is calculated as follows:

$$\mathbf{EF_{grid,CM,y} = EF_{grid,OM,y} \times W_{OM} + EF_{grid,BM,y} \times W_{BM}}$$

Where:

$EF_{grid,CM,y}$ = CM CO_2 emission factor in year y (tCO_2/MWh)

$EF_{grid,OM,y}$ = OM CO_2 emission factor in year y (tCO_2/MWh)

$EF_{grid,BM,y}$ = BM CO_2 emission factor in year y (tCO_2/MWh)

W_{OM} = Weighting of OM emissions factor

W_{BM} = Weighting of BM emissions factor

According to the Tool, for solar power generation projects, the weights to be applied are: W_{OM} = 0.75 and W_{BM} = 0.25 (owing to their intermittent and non-dispatchable nature) for the first crediting period and for subsequent crediting periods.

Parameter	Unit	Description	Applicable project types	Applicable values (First, Second and Third Crediting Period)
$EF_{grid, CM, y}$	tCO_2/MWh	Combined margin CO_2 emission factor for the SENI	Wind and solar power generation project activities	0.6216

According to this the baseline emissions are calculated as follows:

$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$ = 151,262 MWh x 0.6216 tCO_2/MWh

BE_y = 94,024 tCO_2

<u>Project Emissions</u>

As per the methodology, project emissions in year y (PE_y) for solar power projects are null. Thus, $PE_y = 0$.

<u>Leakage</u>

As per the methodology, leakage emissions in year y (LE_y) are null. Thus, $LE_y = 0$.

B.6.2 Data and parameters fixed ex ante

According to the current methodology for the calculation of the ex-ante emission reductions, it is necessary to calculate the emission factor of the electric grid; as per the paragraph above. This parameter is not monitored during the crediting period but has been determined before the design certification and will remain fixed during the crediting period:

SDG 13

Data/parameter	$EF_{grid,CM,y}$
Unit	tCO_2/MWh
Description	Combined margin CO_2 emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system"
Source of data	As per the document ASB0047-2020 "Standarized baseline Grid Emission Factor for the Dominican Republic" (V 01.0).
Value(s) applied	0.6216 tCO_2/MWh
Choice of data or Measurement methods and procedures	The choice is made as per the most recent "Tool to calculate the emission factor for an electricity system"
Purpose of data	Baseline emissions
Additional comment	

B.6.3 Ex ante estimation of SDG Impact

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SDG 7: Affordable and Clean Energy – The project will generate 151,262 MWh/yr.

SDG 8: Decent Work and Economic Growth – During the Construction phase, the project provided around 500 jobs to the locals while during the Operation and Maintenance phase, 100 job opportunities will be made available to the local people. Equal opportunities in employment will be available for the local community, regardless of discrimination on any basis.

SDG 9: Industry, Innovation, and Infrastructure - This project will contribute to the energy mix with renewable energy that industries will use to make industrial processes sustainable. If more renewable energy plants are introduced less non-renewable resources like fossil fuels and gas will be used. As more renewable energy plants are introduced into the Dominican energy mix, industries and processes will use this clean energy to produce and therefore industrial processes will become more sustainable. It is estimated that annually 3 new renewable energy projects will be introduced into energy mix.

SDG 13: Climate Action -

Net GHG Emission Reductions and Removals

As explained above in the text, emission reductions are calculated as follows:

$$\mathbf{ER_y = EG_{facility,y} = EF_{grid,CM,y}}$$

Where:

ER_y = Emission reductions in year y (tCO_2)

$EG_{facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh)

$EF_{grid,CM,y}$ = CM CO_2 emission factor in year y (tCO_2/MWh)

For the ex-ante calculation of emission reductions, the quantity of net electricity supplied to the grid is estimated based on solar monitoring carried out in the region of the project. The proposed project activity is expected to supply an average of 151.2 GWh per year to the grid.

As shown above, the CM emission factor is calculated as follows:

$$\mathbf{EF_{grid,CM,y} = EF_{grid,OM,y} \times W_{OM} + EF_{grid,BM,y} \times W_{BM}}$$

Where:

$EF_{grid,CM,y}$ = CM CO_2 emission factor in year y (tCO_2/MWh)

$EF_{grid,OM,y}$ = OM CO_2 emission factor in year y (tCO_2/MWh)

$EF_{grid,BM,y}$ = BM CO_2 emission factor in year y (tCO_2/MWh)

W_{OM} = Weighting of OM emissions factor

W_{BM} = Weighting of BM emissions factor

According to the Tool, for solar power generation projects, the weights to be applied are: W_{OM} = 0.75 and W_{BM} = 0.25 (owing to their intermittent and non-dispatchable nature) for the first crediting period and for subsequent crediting periods.

Therefore, the annual value of emission reductions estimated for the project activity is the following:

ER_y = 151,262 MWh x 0.6216 tCO_2/MWh = 94,024.45 tCO_2

Approximate values for Pedro Corto Solar are presented below in the Table.

Table. Values of annual emission reductions for Pedro Corto Solar

Year	Estimated baseline emissions or removals (tCO_{2e})	Estimated project emissions or removals (tCO_{2e})	Estimated leakage emissions (tCO_{2e})	Estimated GHG emission reductions or removals (tCO_{2e})
June to December 2025	47,012	0	0	47,012
2026	94,024	0	0	94,024
2027	94,024	0	0	94,024
2028	94,024	0	0	94,024
2029	94,024	0	0	94,024
January to May 2030	47,012	0	0	94,024
Total	470,120	0	0	470,120

B.6.4 Summary of ex ante estimates of each SDG Impact

SDG 7: Affordable and Clean Energy

Year	Baseline estimate	Project estimate	Net benefit
June to December 2025	0	75,631 MWh	75,631 MWh
Year 2026	0	151,262 MWh	151,262 MWh
Year 2027	0	151,262 MWh	151,262 MWh
Year 2028	0	151,262 MWh	151,262 MWh
Year 2029	0	151,262 MWh	151,262 MWh
January to May Year 2030	0	75,631 MWh	75,631 MWh
Total	**0**	**756,310 MWh**	**756,310 MWh**
Total number of crediting years 5			
Annual average over the crediting period	151,262 MWh		

SDG 8: Decent Work and Economic Growth

During the Construction phase, the project provided around 500 jobs to the locals while during the Operation and Maintenance phase, 100 job opportunities will be made available to the local people. Equal opportunities in employment will be available for the local community, regardless of discrimination on any basis. The positions needed for the project will be open equally to men and women. All staff to be employed for a specific task, will also need in addition special training and education in order to successfully perform the operational activities of this new technology. It is very important to emphasize that even though, the Dominican Republic does not have a gender policy that applies to this specific type of project, the project developer seeks to fulfill a gender-sensitive design and implementation, both for the construction phase and the operation phase. Priority will be given to local people to apply to the different positions.

Year	Baseline estimate	Project estimate	Net benefit
June to December 2025	0	100 jobs	100 jobs
Year 2026	0	100 jobs	100 jobs
Year 2027	0	100 jobs	100 jobs
Year 2028	0	100 jobs	100 jobs
Year 2029	0	100 jobs	100 jobs
January to May Year 2030	0	100 jobs	100 jobs
Total	**0**	**100**	**100**
Total number of crediting years 5			
Total number of jobs created over the crediting period	100 jobs		

SDG 9: Industry, Innovation and Infrastructure
This project will contribute to the energy mix with renewable energy that industries will use to make industrial processes sustainable. If more renewable energy plants are introduced less non-renewable resources like fossil fuels and gas will be used. As more renewable energy plants are introduced into the Dominican energy mix, industries and processes will use this clean energy to produce and therefore industrial processes will become more sustainable. It is estimated that annually 3 new renewable energy projects will be introduced into energy mix.

Year	Baseline estimate	Project estimate	Net benefit
June to December 2025	0	3 projects	3 projects
Year 2026	0	3 projects	3 projects
Year 2027	0	3 projects	3 projects
Year 2028	0	3 projects	3 projects
Year 2029	0	3 projects	3 projects

January to May Year 2030	0	3 projects	3 projects
Total	**0**	**15 projects**	**15 projects**
Total number of crediting years 5			
Annual average over the crediting period	3 projects		

SDG 13: Climate Action

Year	Baseline estimate	Project estimate	Net benefit
June to December 2025	47,012	0	47,012
Year 2026	94,024	0	94,024
Year 2027	94,024	0	94,024
Year 2028	94,024	0	94,024
Year 2029	94,024	0	94,024
January to May Year 2030	47,012	0	47,012
Total	**470,120**	**0**	**470,120**
Total number of crediting years 5			
Annual average over the crediting period	94,024		

B.7. Monitoring plan

B.7.1 Data and parameters to be monitored

SDG 7 SDG 13

Data / Parameter	$EG_{facility,y}$
Unit	MWh/year
Description	Quantity of net electricity generation supplied by the project plant to the grid in year y.
Source of data	Monthly reports from Organismo Coordinador del Sistema Eléctrico Nacional OC-SENI.

Value(s) applied	151,262 MWh
Measurement methods and procedures	Sum total amount of renewable energy
Monitoring frequency	Annual
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	To monitor SDG 7
Additional comment	The values applied are based on the calculations made above in the document.

SDG 8

Data / Parameter	Employment Generation
Unit	Employees
Description	During the construction phase approx. 500 jobs will be made available to locals and in the operation and maintenance phase approx. 100. Employment will be subject to requirements and qualification of the candidates.
Source of data	Employment contracts and Human Resources data.
Value(s) applied	Construction Phase: 500 jobs (Approx.) Operation and Maintenance Phase: 100 jobs (Approx.)
Measurement methods and procedures	Employment generation will be monitored through contracts and Human Resources data.
Monitoring frequency	Once in a monitoring period.
QA/QC procedures	Contracts will be made in compliance to relevant Dominican Law.
Purpose of data	To monitor SDG 8
Additional comment	NA

SDG 9

Data / Parameter	Number of Renewable Energy Projects added to the grid
Unit	Renewable Energy Projects
Description	If more renewable energy plants are introduced less non-renewable resources like fossil fuels and gas will be used. As more renewable energy plants are introduced into the Dominican energy mix, industries

	and processes will use this clean energy to produce and therefore industrial processes will become more sustainable.
Source of data	Anual report from Organismo Coordinador del Sistema Eléctrico Nacional OC-SENI.
Value(s) applied	3 new projects every year
Measurement methods and procedures	New projects will be monitored through the data of the Annual Report from OC-SENI.
Monitoring frequency	Once in a monitoring period.
QA/QC procedures	Data will be verified with the official data of electricity generation in the grid.
Purpose of data	To monitor SDG 9
Additional comment	NA

SDG 13

Data / Parameter	$ER_{,y}$
Unit	tCO_2
Description	Emission reductions
Source of data	Calculated based on information provided by the grid operator - Organismo Coordinador del Sistema Eléctrico Nacional OC-SENI and others shown in the document Standardized baseline Grid Emission Factor for the Dominican Republic Version 01.0
Value(s) applied	94,024 tCO2
Measurement methods and procedures	Calculated as per ACM 0002 Methodology Version 21.0
Monitoring frequency	Every verification period
QA/QC procedures	Refer to $EG_{facility,y}$
Purpose of data	To monitor SDG 13
Additional comment	The values applied are based on the calculations made above in the document.

B.7.2 Sampling plan

>>

NA

B.7.3 Other elements of monitoring plan

>>

The Monitoring Plan basically consists of the procedures to meter the project's electricity generation. Since this is the only parameter to be measured and the emission factor is already set ex ante, the process is pretty straightforward. The only variable that needs to be verified and monitored during the total crediting period is the total amount of electricity delivered to the grid.

Monitoring tasks must be implemented according to the monitoring plan in order to ensure that the real, measurable and long-term greenhouse gas (GHG) emission reduction for the proposed project is monitored and reported.

As described in the above section A.3 of Technologies and/or Measures, the precise equipment to be installed for measuring the total amount energy delivered to the grid may be subject to change. The technical aspects of the equipment to be used will be available once the final agreement between the Project owner and the Superintendencia de Electricidad (Superintendency of Electricity) are finished. Nevertheless, the main characteristics that the meter will have for measuring the total amount of energy of electricity delivered to the electric grid at the Point of Common Coupling (PCC), are as follows:

1. Electronic multifunction for being assembled in electric board.
2. Back connection
3. Anti-dust seal
4. Application of a multi tariff billing system
5. Real time measuring access
6. Bi-directional measurement capability
7. Remote data transmission capability
8. Massive memory storage
9. Precision 0.2%

On the other hand, the equipment installed onsite will have the indication of maximum supply, expressed in kWh for the daily double tariff measuring system, with integration periods of 15 minutes. All meters installed will be a "plug-in" type and with the approximate dimensions of 200 mm x 200 mm.

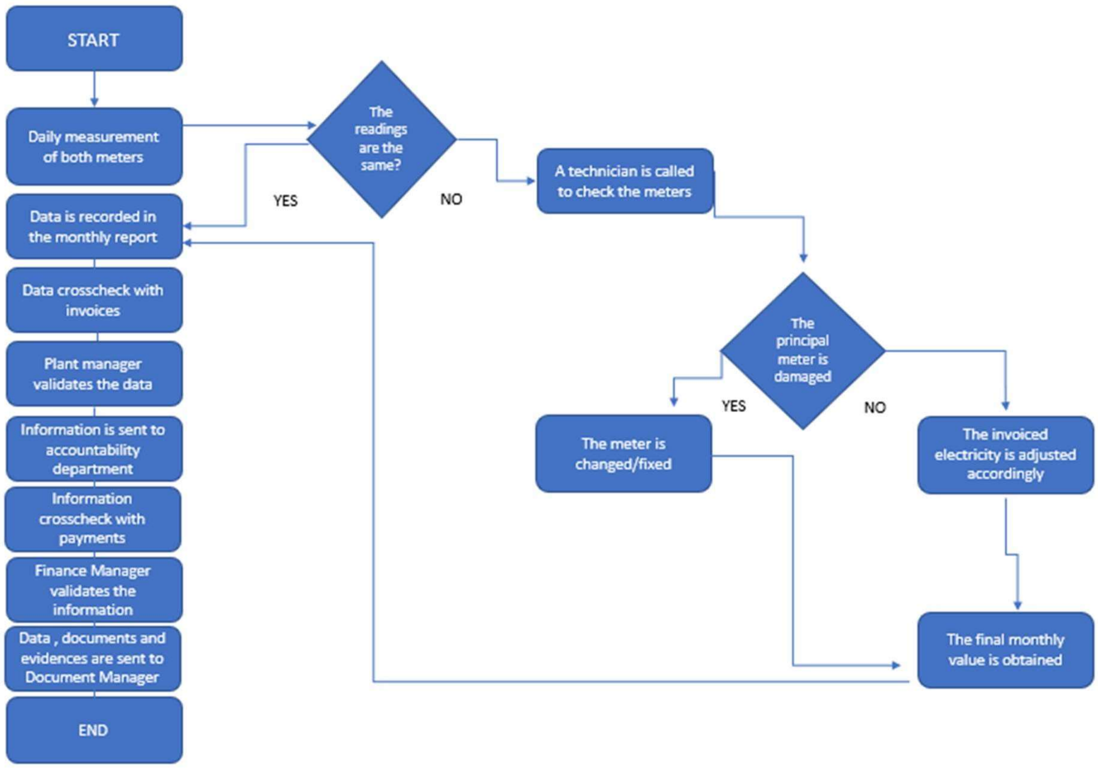
All the equipment (meters) will be designed to work in a four wire three phases system.

The nominal tension and the type of service will be self-programmable within the range of 57 – 277 V and will have an operation rage of +- 15% of the nominal tension, base type, with quartz controlled internal digital clock, independent from the frequency of the grid. The multifunction system will be able to register currents and tensions, power factor, active, reactive and apparent power. It will be able to transmit all the data remotely. It will have two communication buses RS 485, one of them commutable to a RS 232, with an internal modem of 33.6 kbps.

Also, the emission reductions of the project will be accounted as the total electricity delivered to the grid, multiplied by the emission factor of the SENI. In order to monitor the total emission reductions a simplified calculation model will be used.
To monitor the output of the electricity of the solar plant two meters will be installed, one for internal control and the other for delivering the electricity to the grid. The installed meters will be sending all the information to the control cabin. All the information will be recorded and stored. The final crosscheck of the data will be with the sales receipts. Also, each one of the structures will have a direct monitoring for detecting any kind of failure in the electric system. All data monitored and required for verification and issuance of VERs will be kept for two years after the end of the last crediting period or the last issuance of VERs, whichever occurs latest.

The roles and responsibilities of the monitoring team will be set after commissioning of the project. The monitoring team will be responsible for measuring, collecting, monitoring, reporting, operating, and maintaining the metering system and other devices. The monitoring team will include workers of the project who will be trained with the proper skills to ensure an adequate operation and maintenance of the solar power plant.

Based on other projects' monitoring procedure from Project Participant, Soventix Caribbean SRL, the following monitoring procedure has been developed to generate the necessary information for the verification phase of the project.



SECTION C. DURATION AND CREDITING PERIOD

C.1. Duration of project

C.1.1 Start date of project

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As per the definition found in the Principles and Requirements document, the Project start date is the earliest date on which the Project Developer has committed to expenditures related to the implementation of the Project. Therefore, the starting date of the project is defined by the purchase agreement for key equipment (in this case solar panels and structure) and according to this the expected start date of the project is 01/03/2024.

C.1.2 Expected operational lifetime of project
>>
25 years

C.2. Crediting period of project

C.2.1 Start date of crediting period

>>

01/06/2025

C.2.2 Total length of crediting period

>>

5 years, renewable twice, therefore 15 years in total.

SECTION D. SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE ASSESSMENT

D.1 Safeguarding Principles that will be monitored

A completed Safeguarding Principles Assessment is in Appendix 1, ongoing monitoring is summarized below.

Principles	Mitigation Measures added to the Monitoring Plan
Principle 3 Labor Rights	Monitoring Parameter: Safety equipment
Principle 3 Negative Economic Consequences	Monitoring Parameter: Number of Local people employed
Principle 4 Energy Supply	Monitoring Parameter: Quantity of net electricity generation supplied by the project to the grid in a year
Principle 4 Harvesting Of Forests	Monitoring Parameter: Relocation and compensation of removed wild flora

D.2. Assessment that project complies with GS4GG Gender Sensitive requirements.

Question 1 - Explain how the project reflects the key issues and requirements of Gender Sensitive design and implementation as outlined in the Gender Policy?	All employees, regardless of gender, will be hired in accordance with the principles of gender equality and transparency established by the Dominican Republic and it will allow men and women to participate in the project

	decisions/design. The positions needed for the project will be open equally to men and women. As per gender policy, all staff to be employed for a specific task, will also need in addition special training and education in order to successfully perform the operational activities of this new technology. It is very important to emphasize that even though, the Dominican Republic does not have a gender policy that applies to this specific type of project, the project developer seeks to fulfill a gender-sensitive design and implementation, both for the construction phase and the operation phase. Priority will be given to local people to apply to the different positions.
Question 2 - Explain how the project aligns with existing country policies, strategies and best practices	Yes, the project complies with the constitution of the country, as well as the laws of the energy sector and others that apply, such as the environmental law and tax code. Also, the project follows the guidelines of the National Gender Equality and Equity Plan (PLANEG) 2020-2030, that is based on four strategic pillars: (a) gender equality from a human rights perspective; (b) mainstreaming, targeting and high impact intervention; (c) institutional mechanisms for ensuring coordination and linkages during the execution process; (d) strengthening the oversight role of the Ministry of Women.

Question 3 - Is an Expert required for the Gender Safeguarding Principles & Requirements?	No, as shown below in the Appendix of this document all of the questions addressed in the Gold Standard Safeguarding Principles & Requirements have been answered with a justification and a mitigation measure if needed. As shown in the table below all of the questions were answered in compliance with local and national laws of the Dominican Republic and with the guidelines and policies that Irradiasol Dominicana SRL and project partner Soventix Caribbean SRL possess. All of the questions of Human Rights, Gender Equality and Human Rights, Community, health and safety working conditions, Sites of cultural and historical heritage, Forced eviction and displacement, Land tenure and other rights, Indigenous people, Corruption, Labour rights, Negative economic consequences, Emissions, Energy supply, Impact on natural water patterns and flow, Erosion and/or water body stability, Landscape modification and soil, Vulnerability to natural disaster, Genetic resources, Release of pollutants, Hazardous and non-hazardous waste, Pesticides and fertilizers, Harvesting of forests, Food, Animal husbandry, High conservation value areas and critical habitats and Endangered species have been answered according to the information and data provided for the development of this

	project and the result of the Environmental Impact study that has been done and this study allows the project to manage different situations that arise during the design, construction and operation phase.
Question 4 - Is an Expert required to assist with Gender issues at the Stakeholder Consultation?	No, the project has applied the Gold Standard Stakeholder Consultation & Engagement Procedure Requirements with Gender issues to conduct its meeting. Due to COVID-19 restrictions the consultation process has been difficult but all measures to ensure the safety of all participants were taken into account. The Dominican Republic has gone through different curfew times along the COVID-19 emergency and therefore the Stakeholder Consultation has posed to be a challenge. The first consultation took place in December 2020 and all invitations were sent 30 days prior to the meeting and the group invited included local people, communities and representatives, stakeholder with land-tenure rights, local policy makers and representatives of local authorities, national government officials, NGOs, a Gold Standard Representative and WWF and IUCN. All invitations were sent via email and post letter where it applied, and an informative poster of the meeting was placed in the local government building. There is evidence of the invitations sent

	that can be found in the Stakeholder Consultation Report. Information of the project has been available to all participants and can be found in a PowerPoint presentation. Due to COVID-19 restrictions the second Stakeholder´s Consultation was held virtually and all of the participants have the contact information of the technical contact and can write or call her in order to pose questions or give input. Also, socialmedia has been made available for stakeholders to contact Irradiasol Dominicana SRL and project partner Soventix Caribbean SRL to maintain an open communication throughout this process. This open communication will be available to all stakeholders throughout the duration of the project.

SECTION E. SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

The below is a summary of the 2 step GS4GG Consultation for monitoring purposes. Please refer to the separate Stakeholder Consultation Report for a complete report on the initial consultation and stakeholder feedback round.

E.1 Summary of stakeholder mitigation measures

>>

Pedro Corto Solar has done the consultation process in accordance with Stakeholder Procedure Requirements and Guidelines as described above. Due to the COVID-19 restrictions, the consultation was made on the same date as the Environmental Impact Declaration – DIA. Since then, the project has carried out a series of activities to inform and consider the communities in the area of direct and indirect influence. The meeting was held on December 1st, 2020 in Pedro Corto´s City Hall.

The general objective of this consultation is to present in detail the facts of the project, as the design, construction and operational activities; as well as the environmental and social aspects, and the Environmental Management Program in order to let the Local Stakeholders know the project as a whole and furthermore, receive their personal comments and questions. Consequently, the project participants were able to solve their questions and discuss them in detail, to clear any doubts. It is important to point that all questions and suggestions there discussed, were written in the DIA report (found as an attachment to this document), to be known and evaluated by the Environment and Natural Resource Ministry. After this process, the Ministry decides if the project follows all the legislations and environmental rules and is viable to obtain the environmental license.

These consultation processes are structured within the Dominican Republic Legislation - Law 64-00.

Due to COVID-19 restrictions, the second consultation was made virtually and an email, phone number and social media was made available for the next two months so stakeholders can send or write any questions or comments. This open line of communication will be available for the community throughout the duration of the project.

Below is a summary of the key comments received during the consultation process, by whom the question was made, and a summary of the answer given by the Project participants.

Sr. José Manuel Jiménez

Question: He is concerned about lead in batteries and what will happen with the batteries and also why does the project have a life of 25 years.

Answer: For his two concerns, Soventix Caribbean answered that the project has a life of 25 years since the solar panel after 25 years production guarantee and the Concession of the project has 25 years.

To answer his question about batteries, even though this project does not contemplate the use of batteries, Soventix Caribbean commented that batteries are kept in a closed container, and they are never opened or the internal material is extracted so this hazardous material does not come in contact with the land. As can be seen in the DIA, a specialized company is in charge of the handling of hazardous material.

Sr. Ariel Zoquier. President of the Sociedad Ecológica de San Juan.

Question: He is concerned about the corporate social responsibility and how the project participants will manage this, also if the mayor´s office is going to construct another road for transport of material and machinery and if there is going to be a medical facility for the workers of the project.

Answer: For his concern about the medical facility, the project by law is required to have a medical facility inside the project if there are more than 50 people working in site. For his concern about the road, the mayor´s office cannot construct a road, but the project management will try to spread the flow of vehicles and machinery throughout the day in order to avoid traffic overcrowding. For his concern on corporate social responsibility, Soventix Caribbean made it very clear that there is an open channel communication between the project participants, community, local authorities and everybody interested in this project.

Sr. Julio de la Rosa, Evangelical pastor and representative of this church.

Question: He had more of a comment rather than a question on the corporate social responsibility of the project. After listening to the presentation in which he saw the positive and negative impacts of the project, at the end, he feels that there are more positive impacts. He feels that this project will positively impact the region and the community, and he is very hopeful.

Sr. Manuel Feliz, Mayor

Question: He is concerned with once the Project obtains all the permits how long it will take to begin construction and what is the cost of the project.

Answer: For his concern on the start date of the project, the project participants have 6 months by law to begin construction once all the permits are granted. The cost of the project is still an estimate since solar panels prices and solar technology varies very fast, that is why there is an estimate, and the Ministry of Environment has an estimate in order to award the environmental license.

Víctor Ramírez Peralta, President of the Neighborhood Association of El Paraíso, Pedro Corto

Question: He had more of a comment rather than a question, and his comment was about the benefits that the project would bring to the region and the community. He acknowledges that this type of renewable energy project brings more benefits to the community not only as more electricity but also with less negative impacts to the environment. He asks the local authorities to include them in meetings to discuss different topics and he is grateful for the open communication channel that the project participants have with all the actors that are part of this project. He believes that this communication must be open throughout the project in order to avoid environmental damages on the long run.

E.2 Final continuous input / grievance mechanism

Method	Include all details of Chosen Method (s) so that they may be understood and, where relevant, used by readers.
Continuous Input / Grievance Expression Process Book (mandatory)	The chosen method was email, phone number and social media. Due to COVID-19 restrictions, throughout the country physical meetings have been restricted and the majority of people are working remotely. m.alfau@soventix.energy +1 (809) 540 7828 Instagram: @soventix.energy

Facebook: Soventix Caribbean

GS Contact (mandatory)	help@goldstandard.org
Other	

APPENDIX 1 - SAFEGUARDING PRINCIPLES ASSESSMENT

Complete the Assessment below and copy all Mitigation Measures for each Principle into SECTION D above. Please refer to the instructions in the Guide to Completing this Form.

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, managementor risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
Principle 1. Human Rights			
1. The Project Developer and the Project shall respect internationally proclaimed human rights and shall not be complicit in violence or	1. No	1. The project respects human rights and is not including in any way the violence or human rights abuses of the communities' activities next to the Project implementation site; neither to the urban zones close to the site. The Project will be developed in an area which is not inhabited and has no commercial or urban use. All the people connected to the grid in Pedro Corto will receive renewable energy.	1. Not needed.

human rights abuses of any kind as defined in the Universal Declaration of Human Rights 2. The Project shall not discriminate with regards to participation and inclusion	2. No	2. The project is not in any way discriminatory regarding gender, race, religion etc. The employees will be hired in accordance with the principles of gender equality and transparency established by Dominican Republic law. Besides, households in Pedro Corto without discrimination will receive renewable energy.	2. Not needed.
Principle 2. Gender Equality			
1. The Project shall not directly or indirectly lead to/contribute to adverse impacts on gender	1. No 2. No 3. No	1. The project does not impact or put at risk women´s access to or control of resources, entitlements and benefits. The Project will be developed in an area which is not inhabited and has no commercial or urban use. **2.** All employees men and women will be hired in accordance with the principles of gender equality and transparency established by the law in the Dominican Republic and allow men and women to participate in the project decisions/design. Irradiasol Dominicana SRL currently is partnering with Project Developer - Soventix Caribbean SRL - and the latter has a clear hiring policy in which it states that it will not carry out any type of gender discrimination for the hiring nor in the salary allocation**.**	1. Not needed. 2. Not needed.

equality and/or the situation of women 2. Projects shall apply the principles of non-discrimination, equal treatment, and equal pay for equal work 3. The Project shall refer to the country's national gender strategy or equivalent national commitment to aid in assessing	4. No	3. There is a National Gender Equality and Equity Plan (PLANEG III) 2020-2030 that is based on four strategic pillars: (a) gender equality from a human rights perspective; (b) mainstreaming, targeting and high impact intervention; (c) institutional mechanisms for ensuring coordination and linkages during the execution process; (d) strengthening the oversight role of the Ministry of Women.https://mujer.gob.do/transparencia/phocadownload/Publicaciones/Planeg/PLANEG%20III%202020-2030.pdf Irradiasol Dominicana SRL and project partner – Soventix Caribbean SRL, comply with all Dominican legislation. There is no gender discrimination for the hiring nor in the salary allocation. 4. NA	3. Not needed. 4. Not needed.

gender risks 4. (where required) Summary of opinions and recommendations of an Expert Stakeholder(s)			
Principle 3. Community Health, Safety and Working Conditions			
1. The Project shall avoid community exposure to increased health risks and shall not adversely affect the health of the workers	1. No	1. Employees will receive a proper social insurance and proper work protection clothes. Also, the employees receive regular capacity building whenever needed.	1. Not needed.

and the community			
Principle 4.1 Sites of Cultural and Historical Heritage			
Does the Project Area include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture? >>	No	The Project does not involve the degradation of the surrounding project area, natural habitats and does not have any permanent negative impact on the region's biodiversity. The project area does not include such sites, structures or objects. **Note:** The land is not a palm farm and does not have farming activities. As the EIA states the land characteristics are not deemed for agriculture. The area surrounding the project will not have any permanent impact.	Not needed.
Principle 4.2 Forced Eviction and Displacement			
Does the Project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)?	No	The project does not require or cause relocation of people as it will be developed in an area which is not inhabited and has no commercial or urban use.	NA

>>			
Principle 4.3 Land Tenure and Other Rights			
a. Does the Project require any change, or have any uncertainties related to land tenure arrangements and/or access rights, usage rights or land ownership? b. For Projects involving land use tenure, are there any uncertainties with regards to land tenure, access rights, usage rights or land ownership?	a. No b. No	a. The project does not change or influence land tenure arrangements and/or rights. The project is located in an isolated area and strictly respects all the local regulations regarding the development of this type of projects and activities. The owner of the land where the project will be constructed is Agroindustria del Sur, SRL, and at this moment the land is leased through a contract to Irradiasol Dominicana SRL. Project participant Soventix Caribbean SRL has the right to buy the land once the definite concession for the project is obtained. All of this is documented in a contract that can be found as an attachment. b. The project does not involve land tenure. It will be developed in an area which is not inhabited and has no commercial or urban use. The owner of the land where the project will be constructed is Agroindustria del Sur, SRL, and at this moment the land is leased through a contract to Irradiasol Dominicana SRL. Project participant Soventix Caribbean SRL has the right to buy the land once the definite concession for the project is obtained. All of this is documented in a contract that can be found as an attachment.	a. Not needed. b. Not needed.
>>			
Principle 4.4 - Indigenous people			
Are indigenous peoples present	No	The Project activity is not including in any way the irruption of the indigenous communities' activities next to the Project implementation site; neither to the urban	Not needed.

in or within the area of influence of the Project and/or is the Project located on land/territory claimed by indigenous peoples? >>		zones close to the site. The employees who will be participating in the construction and operation stages will be protected by the local and national legal framework.	
Principle 5. Corruption			
1. The Project shall not involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects	No	The Project does not involve and is not complicit in corruption. The complete development of the Project, for example technical, legal and economic development, is done in strict respect of the local and national law. Soventix Caribbean SRL, project partner of Irradiasol Dominicana SRL, is compliant with current local laws on corruption (for example, Money Laundering, etc).	Not needed.
Principle 6.1 Labour Rights			

1. The Project Developer shall ensure that all employment is in compliance with national labour occupational health and safety laws and with the principles and standards embodied in the ILO fundamental conventions 2. Workers shall be able to	1. No 2. No 3. No	1. All the people who will be employed during the construction and operation of the Project, will be hired in accordance with the legal framework established by the Dominican Republic national labour laws. 2. The project does not restrict employees of their rights and freedom. The Project respects and will respect the freedom of association of the employees and will be developed in strict respect of the local and national laws of the Dominican Republic.	1. Not needed. 2. Not needed.

establish and join labour organisations 3. Working agreements with all individual workers shall be documented and implemented and include: a) Working hours (must not exceed 48 hours per week on a regular basis), AND	4. No 5. No	3. Soventix Caribbean SRL, project partner of Irradiasol Dominicana SRL, has agreements and job profiles in their current operating projects that contain all the requested criteria by Gold Standard. Their contracts contain the following: a. 42 hours per week b. Duties and tasks are listed in the job profile. c. Salary is defined in the agreement (internal employee regulation). d. Modalities of health insurance, etc. are described in the agreement. e. Modalities of termination of contract and resignation are described in the agreement. f. Annual leave is 14 days. 4. All employees are and will be adults (18 years old onwards), men and women, and they will be hired in accordance with the principles of gender equality and transparency established by Dominican Republic law. No children are/or will be employed by Irradiasol Dominicana SRL and its project partner Soventix Caribbean SRL. 5. The Project provides workers with a safe and healthy work environment and the work environment will be regulated by the national laws of security and hygiene established in the Dominican Republic. On the other hand, all workers will be provided the necessary training needed in order to be able to carry out their tasks in an environment governed by the highest security standards. When necessary and reasonable, workers employed will have to undergo and pass health tests, and they will also be provided with the necessary security equipment in	3. Not needed. 4. Not needed. 5. Monitoring parameter:

b) Duties and tasks, AND c) Remuneration (must include provision for payment of overtime), AND d) Modalities on health insurance, AND e) Modalities on termination of the contract with provision for voluntary		order to carry out their jobs. Where and when needed, employees will also receive the adequate and sufficient training needed in order to carry out their responsibilities.	Safety equipment .

resignation by employee, AND f) Provision for annual leave of not less than 10 days per year, not including sick and casual leave. 4. No child labour is allowed (Exceptions for children working on their families' property requires an [Expert](#)			

Stakeholder opinion) 5. The Project Developer shall ensure the use of appropriate equipment, training of workers, documentation and reporting of accidents and incidents, and emergency preparedness and response measures			
Principle 6.2 Negative Economic Consequences			

1. Does the project cause negative economic consequences during and after project implementation? >>	Yes	The GS income revenues will contribute to the necessary financial stability to obtain the equity investors and the debt financing from the banks involved. The GS revenues reduce the financial and investment barriers (access to bank loans) and will help to increase the acceptance among the lenders and investors of taking additional technical risks of such a new technology applied in the Dominican Republic. Without the GS financial stability, the project activity will be hard to implement due to its prohibitive high investment costs, if compared to other, simpler investment alternatives in the electricity sector of the Dominican Republic. The project does not affect the local economy negatively. On the contrary, the project generates local added value through creation of jobs and income. The project will create new and permanent employment that will be available to anyone with the proper skills to do the job, without considering age, gender, race or culture. Also, the Project proposes the continuous technical training of the staff and the collaboration with other relevant national and international actors. The Project Activity during the construction stage, will generate around 500 direct jobs and 1.500 indirect jobs, and during the operation and maintenance stage at least 100 positions.	Monitoring Parameter : Number of local people employed.
Principle 7.1 Emissions			
Will the Project increase greenhouse gas emissions over the Baseline Scenario? >>	No	The project reduces GHG emissions over the baseline.	Not needed.
Principle 7.2 Energy Supply			
Will the Project use energy from a local grid or	Yes	The Project does not use energy from a local grid, but will supply clean renewable energy to the Dominican energetic portfolio, and therefore it contributes to decrease	Monitoring Parameter : Quantity

power supply (i.e., not connected to a national or regional grid) or fuel resource (such as wood, biomass) that provides for other local users?		the consumption of fossil fuels. Moreover, long-term it is expected that the generation costs will become cheaper if compared to diesel generators.	of net electricity generation supplied by the project to the grid in a year
>>			
Principle 8.1 Impact on Natural Water Patterns/Flows			
Will the Project affect the natural or pre-existing pattern of watercourses, groundwater and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic	No	The project does not influence water sources. There is no significant impact of the project activity on the water quality or quantity because the plant is not using any kind of water during its operation. Some minor impacts are mentioned in the Environmental Impact Study done before the construction of the site.	Not needed.

connectivity or water scarcity?			
>>			
Principle 8.2 Erosion and/or Water Body Instability			
a. Could the Project directly or indirectly cause additional erosion and/or water body instability or disrupt the natural pattern of erosion? b. Is the Project's area of influence susceptible to excessive erosion and/or water body instability?	a. No b. No	a. The project does not influence water sources. There is no significant impact of the project activity on the water quality or quantity because the plant is not using any kind of water during its operation. Some minor impacts are mentioned in the Environmental Impact Study that has already been done and are taken into account before the construction of the site. b. The project does not cause directly or indirectly erosion or water body instability. Also, the project activity is isolated in a way that there are no external inputs or incoming goods into the project site and the only output is the electricity generated, which is delivered to the national grid.	a. Not needed. b. Not needed.
>>			

Principle 9.1 Landscape Modification and Soil			
Does the Project involve the use of land and soil for production of crops or other products? >>	No	The project does not involve the use of land and soil for production of crops since the land chosen for the project is not deemed for agriculture.	Not needed.
Principle 9.2 Vulnerability to Natural Disaster			
Will the Project be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions? >>	No	The project activity is not susceptible and does not lead to increase vulnerability in any aspects.	Not needed.
Principle 9.3 Genetic Resources			
Could the Project be negatively	No	The project does not involve use of GMO and is not prone to be affected by GMO contamination.	Not needed.

impacted by or involve genetically modified organisms or GMOs (e.g., contamination, collection and/or harvesting, commercial development, or take place in facilities or farms that include GMOs in their processes and production)? >>			
Principle 9.4 Release of pollutants			
Could the Project potentially result in the release of pollutants to the environment? >>	No	The project activity does not result in the release of pollutants. As mentioned before, the project activity only produces electricity. The noise level is kept to a minimum as there are no heavy machineries involved and there are no other pollutants emitted. Residues and wastes have to be managed according to the Environmental Impact Study.	Not needed.
Principle 9.5 Hazardous and Non-hazardous Waste			

Will the Project involve the manufacture, trade, release, and/ or use of hazardous and non-hazardous chemicals and/or materials? >>	No	The project does not manufacture, trade or release hazardous chemicals or materials. Residues and wastes have to be managed according to the Environmental Impact Study.	Not needed.
Principle 9.6 Pesticides & Fertilisers			
Will the Project involve the application of pesticides and/or fertilisers? >>	No	The project does not involve use of pesticides and/or fertilisers as it is a renewable energy project.	Not needed.
Principle 9.7 Harvesting of Forests			
Will the Project involve the harvesting of forests? >>	Yes	During the construction of the project activity there will be a relocation of the wild vegetation that will be removed. Compensation measures will take place to re-establish the biodiversity removed at the project site. Therefore, the long-term impact of the project activity is neutralised. Affected flora is listed in the Environmental Impact Study. In the study area six protected species were found that are included in the Red List of the Vascular Flora of the Dominican Republic*:* ***Lemaireocereus hystrix; Opuntia taylorii; Hylocereus undatus****;* ***Cylindropuntia caribaea and Harrisia divaricata****.* These five species belong to the **Cactaceae family** and are protected by the Appendix II of the Convention on International Trade in Endangered	Monitoring Parameter : Relocation and compensation of removed wild flora

		Species of Wild Fauna and Flora (CITES), also ***Guaiacum oficcinale*** (VU). No species included in the Red List of IUCN were found; nevertheless all these species will be treated accordingly during the construction phase. At least, an equal amount of native flora will be replanted next to the project site and different plant species withinthe boundaries of the project site. Fruits harvesting or the production of any other crop will be desirable but will not be monitored or required as an outcome of the project activity.	
Principle 9.8 Food			
Does the Project modify the quantity or nutritional quality of food available such as through crop regime alteration or export or economic incentives? >>	No	The project does not affect availability or quality of food.	Not needed.
Principle 9.9 Animal husbandry			
Will the Project involve animal husbandry? >>	No	The Project does not involve animal husbandry as it a Solar Power Plant.	Not needed.
Principle 9.10 High Conservation Value Areas and Critical Habitats			

Does the Project physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified? >>	No	No; the Project does not affect or alter the surrounding project area natural habitats and does not have any permanent negative impact on the region's biodiversity.	Not needed.
Principle 9.11 Endangered Species			
a. Are there any endangered species identified as potentially being present within the Project boundary (including those that may route through the area)? b. Does the Project	a. No b. No	a. According to the EIA study, for the management and conservation of flora and fauna species, the relocation of the flora and fauna species protected by current environmental legislation, will be made according to the results and recommendations of the study. This has an effect on the protection of the habitat of the flora species identified in the project area. The maintenance of good coordination with the authorities of the Ministry of the Environment of the area is also foreseen to ensure the process of relocation of the protected species in suitable places and the protection of the habitats of the flora species. According the Environmental Compliance Report (December 2020), measures for the conservation of flora, fauna and vulnerable areas are as follows: 1. Relocation of the protected flora species found in the project area. 2. Increase in vegetation cover with the reforestation of critical areas in the area.	a. Not needed.

potentially impact other areas where endangered species may be present through transboundary affects?		3. Reuse or distribution in the project area of the land extracted from the excavations. b. According to the EIA study there is no transboundary affects, the use of lateral lending areas is not envisaged to be used in the construction phase of the project, given that there are good access roads throughout the project's perimeter.	
>>			

APPENDIX 2- CONTACT INFORMATION OF PROJECT PARTICIPANTS

Organization name	Soventix Caribbean SRL
Registration number with relevant authority	RNC 1-31-20032-1
Street/P.O. Box	Calle Marginal de la Av. Núñez de Cáceres No. 366
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Organization name	Irradiasol Dominicana SRL
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APPENDIX 3- LUF ADDITIONAL INFORMATION

Risk of change to the Project Area during Project Certification Period:	
Risk of change to the Project activities during Project Certification Period:	
Land-use history and current status of Project Area:	
Socio-Economic history:	
Forest management applied (past and future)	
Forest characteristics (including main tree species planted)	
Main social impacts (risks and benefits)	
Main environmental impacts (risks and benefits)	
Financial structure	
Infrastructure (roads/houses etc):	
Water bodies:	
Sites with special significance for indigenous people and local communities - resulting from the Stakeholder Consultation:	
Where indigenous people and local communities are situated:	
Where indigenous people and local communities have legal rights, customary rights or sites with special cultural, ecological, economic, religious or spiritual significance:	

APPENDIX 4-SUMMARY OF APPROVED DESIGN CHANGES

Please refer to Design Change Requirements for more information on procedures governing Design Changes

Revision History

Version	Date	Remarks
1.2	14 October 2020	Hyperlinked section summary to enable quick access to key sections Improved clarity on Key Project Information Inclusion criteria table added Gender sensitive requirements added Prior consideration (1 yr rule) and Ongoing Financial Need added Safeguard Principles Assessment as annex and a new section to include applicable safeguards for clarity Improved Clarity on SDG contribution/SDG Impact term used throughout Clarity on Stakeholder Consultation information required Provision of an accompanying Guide to help the user understand detailed rules and requirements
1.1	24 August 2017	Updated to include section A.8 on 'gender sensitive' requirements
1.0	10 July 2017	Initial adoption